PD-LMI package
User Manual

Version v0.4.0

July 2026

Abstract

This manual teaches the PD-LMI package from the continuous matrix inequality to the finite semidefinite program. Its core representation is a continuous piecewise-polynomial matrix function on a physical cell grid, expressed in a cell-local tensor-product Bernstein basis. A worked 2-by-2 LPV Lyapunov problem first introduces quadratic decisions, physical cells, local Bernstein coefficients, matrix algebra, parameter-rate vertices, certificate selection, and the YALMIP boundary. The reference chapters then document `pdbase`, `pdmat`, `pdvar`, and `pdlmi`. Two self-contained appendices develop the Bernstein and polynomial-positivity background for readers who know basic matrices and calculus but have not previously used Bernstein bases or SOS certificates.

Contents

Release History

v0.4.0 (July 2026). Inequalities now select one global comparison mode from the original coefficient data: square Hermitian families use semidefinite constraints, while every other family uses entry-wise scalar constraints. The release also adds direct coefficient equality through `==`, including rectangular and nonsymmetric expressions, and reorganizes the class-property, plotting, and Bernstein-storage documentation.

1

From a PD-LMI to Solver Constraints

1.1 Begin With The Continuous Inequality

Let the scheduling vector be $\boldsymbol{\rho} = (\rho_1, \dots, \rho_\ell) \in \mathcal{P}$. A *parameter-dependent LMI* is abbreviated *PD-LMI* throughout this manual and is the umbrella problem class named by the package. When a model also uses the bounded rate $\dot{\boldsymbol{\rho}} \in \mathcal{V}$ and derivatives of its decision functions, we call it a *differentiable parameter-dependent LMI*, or *DPD-LMI*. A canonical DPD-LMI is

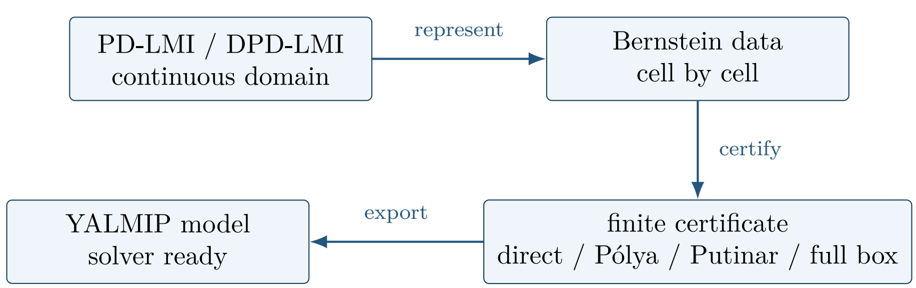
$$\sum_i R_i(\boldsymbol{\rho}) \left(\sum_{j=1}^{\ell} \frac{\partial x_i(\boldsymbol{\rho})}{\partial \rho_j} \dot{\rho}_j \right) + F_0(\boldsymbol{\rho}) + \sum_i F_i(\boldsymbol{\rho}) x_i(\boldsymbol{\rho}) \prec 0.$$

The known functions are the R_i and F_i; the differentiable functions x_i are decisions. After known data are fixed, every term must remain affine in the decision coefficients. Removing the derivative term recovers an ordinary PD-LMI. Thus “DPD-LMI” adds a derivative qualifier; it does not replace the package’s PD-LMI name.

A concrete one-parameter example is the LPV Lyapunov condition

$$\begin{aligned} P(\rho) &\succeq I, \qquad (1) \\ \dot{P}(\rho, \dot{\rho}) + P(\rho)A(\rho) + A(\rho)^{\mathsf{T}} P(\rho) &\preceq -\varepsilon I, \qquad (2) \\ \dot{P}(\rho, \dot{\rho}) &= \frac{\partial P(\rho)}{\partial \rho} \dot{\rho}. \end{aligned}$$

Taking $\dot{\rho} = 0$, or removing $\dot{P}$, gives the familiar PD-LMI Lyapunov inequality. In the worked path below P is 2-by-2, $\rho \in [0, 1]$, $\dot{\rho} \in [-1, 1]$, and $\varepsilon = 10^{-6}$. The package inserts no hidden strictness margin; the displayed εI is part of the user’s model.



The package changes representation and certificate; it does not replace YALMIP’s solver interface.

1.2 Bernstein Cells Support Continuous Lipschitz Matrix Functions

In LPV analysis we often seek a Lyapunov matrix $P(\boldsymbol{\rho})$ that is Lipschitz continuous on the compact parameter set $\mathcal{P}$. Gridding $\mathcal{P}$ makes the model finite and local, but independent cell polynomials would not agree on a grid boundary. The package instead shares the complete Bernstein coefficient face between neighboring cells. Because the trace of a tensor-product Bernstein polynomial on a face depends only on that face's coefficients, this sharing gives C^0 continuity across every cell boundary. Each cell polynomial has bounded first derivatives, and a continuous piecewise-polynomial function on a finite compact grid is therefore globally Lipschitz. Derivatives need not agree across a boundary: the construction guarantees C^0 continuity and Lipschitz regularity, not C^1 continuity.

On one normalized cell, the degree-m Bernstein polynomial basis is

$$B_i^m(\alpha) = \binom{m}{i}(1-\alpha)^{m-i}\alpha^i, \qquad i = 0, \ldots, m,$$

where $\alpha \in [0,1]$ is the forward local coordinate. The basis functions are nonnegative and form a partition of unity. Therefore a matrix polynomial $C(\alpha) = \sum_i B_i^m(\alpha)C[i]$ lies in the convex hull of its Bernstein coefficients. This convex-hull property turns matrix-sign questions into finite sufficient coefficient tests. Exact degree elevation aligns operands without changing their values, tensor-product bases extend the same reasoning to boxes, and separate cell-local storage handles nonuniform physical partitions without inventing global samples.

The coefficients are not power-basis coefficients. For example,

$$p(\alpha) = 1 + 2\alpha + 3\alpha^2 = B_0^2(\alpha) + 2B_1^2(\alpha) + 6B_2^2(\alpha).$$

Thus the degree is two and the Bernstein coefficient row is $[1, 2, 6]$. The basis sum remains one, so elevation can change this row while preserving exactly the same polynomial.

The representation is local even when the physical domain is two-dimensional. The following degree-two matrix example has grid $\{[0, 0.5, 1], [0, 1]\}$, hence two physical cells and nine local coefficient matrices per cell. The public traversal methods return physical-cell rows and zero-based local labels; the accessor and the nested `LocalValues` storage path select the same leaf.

Example

```
grid2 = {[0 .5 1], [0 1]};
A2 = pdmat(grid2, @(rho1,rho2) ...
    [1+rho1+rho2, rho1*rho2; ...
     rho1*rho2,   2+rho1^2], Degree=2);
physicalCells = A2.cells()
localLabels = A2.lbls()
secondCellCoefficients = A2.coeffs([2 1])
sameLeaf = A2.LocalValues{2}{1}
```

Both final variables are the flat nine-entry coefficient cell for physical cell $[2, 1]$; they are not a 3-by-3 grid of point samples. The remaining construction details, exact power-to-Bernstein conversion, and tensor ordering are developed in Appendix A.

1.3 Separate Physical Nodes, Cells, And Local Labels

For one parameter choose physical grid nodes

$$\rho_1^{(1)} < \rho_1^{(2)} < \rho_1^{(3)}.$$

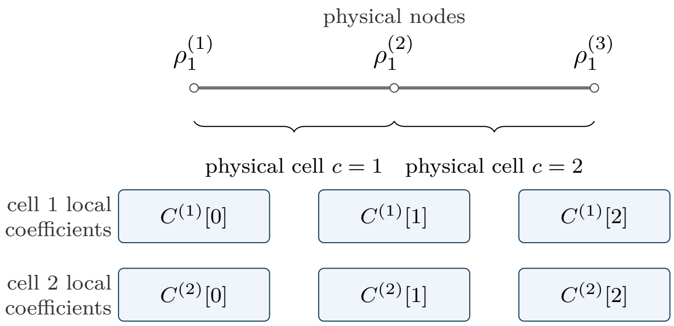
The subscript identifies the parameter direction, while the parenthesized superscript identifies the one-based physical node. The two physical cells have separate identities $c = 1$ and $c = 2$: $[\rho_1^{(1)}, \rho_1^{(2)}]$

and $[\rho_1^{(2)}, \rho_1^{(3)}]$. On cell c, define the forward coordinate

$$\alpha = \frac{\rho_1 - \rho_1^{(c)}}{\rho_1^{(c+1)} - \rho_1^{(c)}} \in [0, 1].$$

For local degree $m = 2$, the three coefficient labels are bracketed local positions, not additional physical nodes:

$$C^{(c)}(\rho_1) = \sum_{i=0}^{2} B_i^2(\alpha) C^{(c)}[i].$$



direction subscripts, parenthesized node indices, cell identity, and local labels have different jobs

In MATLAB, the physical grid and degree are likewise independent choices. The example below uses degree two. Each cell has labels ([0]), ([1]), and ([2]), while adjacent cells share only their common face value. The middle coefficient of each cell is an interior control, not a physical-grid sample.

Example

```
M = pdmat([0 0.5 1], {10, 15, 20, 25, 30}, Degree=2);
physicalCells = M.cells()
localLabels = M.lbls()
physicalCellCount = M.ncell()
coefficientsPerCell = M.ncoeff()
cell1Coefficients = cell2mat(M.coeffs(1))
cell2Coefficients = cell2mat(M.coeffs(2))
```

```
physicalCells =
     1
     2
localLabels =
     0
     1
     2
physicalCellCount =
     2
coefficientsPerCell =
     3
cell1Coefficients =
    10    15    20
cell2Coefficients =
    20    25    30
```

The shared value 20 belongs to the physical interface $\rho_1^{(2)}$. It is seen as $C^{(1)}[2]$ from the left cell and $C^{(2)}[0]$ from the right. This is C^0 face sharing: values meet, but the two cell-local derivatives need not agree. Consequently the representation is a continuous piecewise-polynomial Bernstein representation on a gridding partition, not a B-spline construction and not a claim of global differentiability.

1.4 Store Known Data And Decisions On The Grid

The known LPV matrix in the worked example is affine in ρ. The physical grid has two cells, $[0, 0.5]$ and $[0.5, 1]$. Supplying `Degree=1` asks `pdmat` to retain exact function evaluation and verified cell-local Bernstein coefficient evidence on both cells.

Example

```
yalmip('clear')
grid = {[0 0.5 1]};
A = pdmat(grid, @(rho) [−1, 0.5; −1, −2] + ...
    rho * [−1.3, −20; 2, −10], Degree=1);
matrixSize = size(A)
physicalCellCount = A.ncell()
degree = A.Degree
coefficientsPerCell = A.ncoeff()
A0 = A.evaluate(0)
```

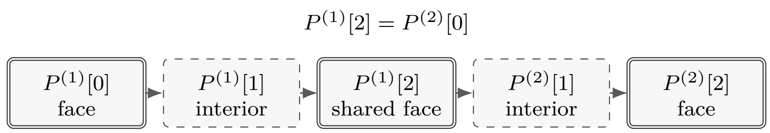
```
matrixSize =
     2     2
physicalCellCount =
     2
degree =
     1
coefficientsPerCell =
     2
A0 =
   -1.0000    0.5000
   -1.0000   -2.0000
```

The decision $P(\rho)$ uses the same physical grid but local degree two. Each cell stores three full symmetric coefficient matrices. The right face of cell 1 and the left face of cell 2 are the same YALMIP decision handle, so the six local positions contain five global control matrices. Only this face is shared; the two interior controls remain independent.

Example

```
P = pdvar(2, grid, "symmetric", Degree=2);
matrixSize = size(P)
degree = P.Degree
isContinuous = P.IsContinuous
localCoefficientsPerCell = numel(P.coeffs(1))
globalControlCount = 2 * P.Degree + 1
```

```
matrixSize =
     2     2
degree =
     2
isContinuous =
  logical
   1
localCoefficientsPerCell =
     3
globalControlCount =
     5
```



six local positions, five global matrix controls; value continuity only

1.5 Form Matrix Algebra In Coefficient Space

Fix a physical cell c. Its identity remains in the parenthesized superscript while the bracketed integers are local Bernstein labels: $P = \sum_i B_i^m P^{(c)}[i]$ and $A = \sum_j B_j^n A^{(c)}[j]$. Their ordered product is

$$PA = \sum_{k=0}^{m+n} B_k^{m+n} R^{(c)}[k], \qquad R^{(c)}[k] = \sum_{i+j=k} \frac{\binom{m}{i}\binom{n}{j}}{\binom{m+n}{k}} P^{(c)}[i] A^{(c)}[j].$$

For the quadratic decision and affine known factor in this worked example, write $P_i = P^{(c)}[i]$ and $A_j = A^{(c)}[j]$. The four cubic coefficients are

$$R^{(c)}[0] = P_0 A_0, \qquad R^{(c)}[1] = \frac{P_0 A_1 + 2P_1 A_0}{3},$$

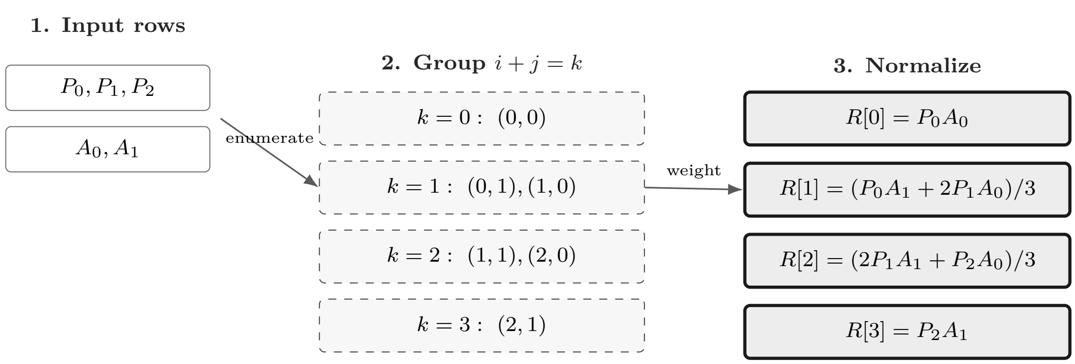
$$R^{(c)}[2] = \frac{2P_1 A_1 + P_2 A_0}{3}, \qquad R^{(c)}[3] = P_2 A_1.$$

The pairs are first grouped by the anti-diagonal relation $k = i + j$, then combined with the binomial weights. This is an ordered convolution, not a samplewise product. Matrix order is preserved in every contribution.

Example

```
PA = P * A;
ATP = A' * P;
degreePA = PA.Degree
degreeATP = ATP.Degree
coefficientCountPA = numel(PA.coeffs(1))
```

```
degreePA =
     3
degreeATP =
     3
coefficientCountPA =
     4
```



The staged view replaces a crossing arrow fan: enumerate ordered products, group by one anti-diagonal, then apply binomial normalization.

1.6 Differentiate And Enumerate Rate Vertices

For a degree-m cell c of width h_c, differentiation is a forward difference of adjacent local coefficients:

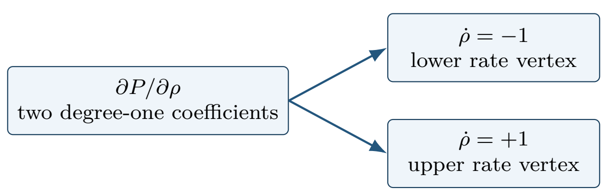
$$\frac{\partial P}{\partial \rho} = \frac{m}{h_c} \sum_{i=0}^{m-1} \left(P^{(c)}[i+1] - P^{(c)}[i]\right) B_i^{m-1}(\alpha).$$

Because $\dot{P} = (\partial P/\partial \rho)\dot{\rho}$ is affine in the rate, checking both vertices -1 and $+1$ covers the whole interval $\dot{\rho} \in [-1, 1]$.

Example

```
D = rhodiff(P, [-1 1]);
derivativeDegree = D.Degree
isContinuous = D.IsContinuous
rateRowCount = size(D.coeffs(1),1)
coefficientsPerRow = size(D.coeffs(1),2)
```

```
derivativeDegree =
     1
isContinuous =
  logical
   0
rateRowCount =
     2
coefficientsPerRow =
     2
```



rate rows are not physical cells and are not extra Bernstein labels

1.7 Compare The Residual And Choose A Certificate

The following MATLAB expressions are a direct transcription of eqs. (1) and (2). On each cell, addition elevates the degree-one derivative rows to the degree-three algebraic residual without changing their represented values.

Example

```
residual = D + P * A + A' * P;
Cdecay = residual <= -1e-6 * eye(2);
Cpositive = P >= eye(2);
residualDegree = residual.Degree
decayConstraintCount = numel(Cdecay.Constraints)
```

```
positivityConstraintCount = numel(Cpositive.Constraints)
```

```
residualDegree =
     3
decayConstraintCount =
    16
positivityConstraintCount =
     6
```

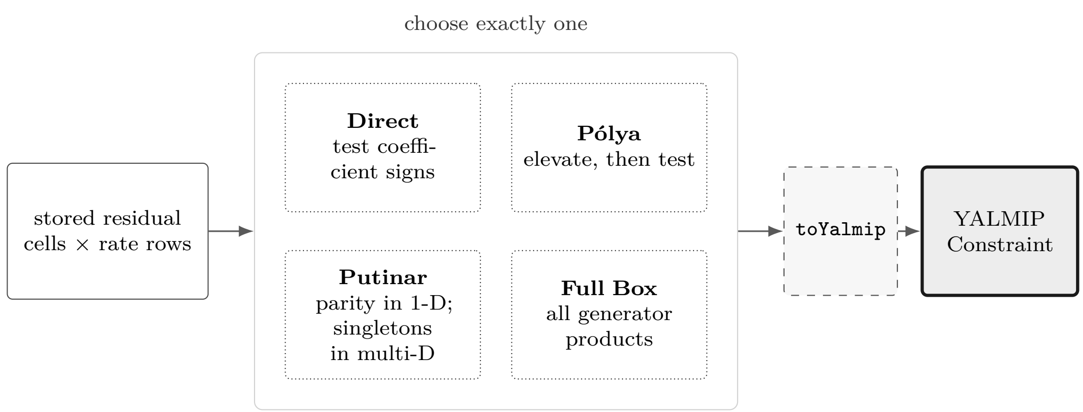
There are two physical cells, two rate vertices, and four degree-three local labels, hence $2 \times 2 \times 4 = 16$ decay constraints. Positivity has no rate rows and uses three degree-two labels on each cell, hence $2 \times 3 = 6$ direct constraints. The shared face is deliberately visited from each physical cell because direct certification is cell-local.

Direct coefficient constraints are the default. Three opt-in alternatives keep the same stored residual but change the finite sufficient certificate:

Example

```
Cpolya = Cdecay.applyPolya(1);
Cputinar = Cdecay.applyPutinar();
Cbox = Cdecay.applyFullBoxPreorder();
directConstraintCount = numel(Cdecay.Constraints)
polyaConstraintCount = numel(Cpolya.Constraints)
putinarConstraintCount = numel(Cputinar.Constraints)
fullBoxConstraintCount = numel(Cbox.Constraints)
```

```
directConstraintCount =
    16
polyaConstraintCount =
    20
putinarConstraintCount =
    28
fullBoxConstraintCount =
    24
```



A failed sufficient certificate does not prove that the continuous inequality is infeasible. Appendix B derives the four implemented choices and places them beside SOS background.

1.8 Export The Finite Model To YALMIP

`toYalmip` returns exactly the stored finite constraints. It does not select a solver, add an objective, or insert a margin.

Example

```
Fdecay = toYalmip(Cdecay);
Fpositive = Cpositive.toYalmip();
decayConstraintCount = length(Fdecay)
positivityConstraintCount = length(Fpositive)
```

```
decayConstraintCount =
    16
positivityConstraintCount =
     6
```

Use ordinary YALMIP calls after that boundary:

Example

```
solver = 'lmilab';
if exist('mosekopt', 'file') ~= 0
    solver = 'mosek';
end
opts = sdpsettings('solver', solver, 'verbose', 0);
sol = optimize([Fdecay, Fpositive], [], opts);
problemCode = sol.problem
```

The final code is produced by the installed solver and should be interpreted with `sol.info`; it is not a package-owned status. The complete tested solver examples in sections 8.2 and 8.3 include assertions appropriate to their solver paths.

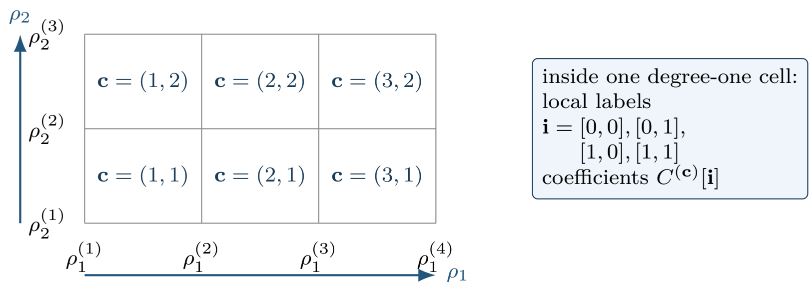
1.9 Generalize From An Interval To A Hypercube

Different parameter directions need not have the same number of physical nodes. For example, four nodes in the first direction and three in the second create a 3-by-2 array of six physical cells:

Example

```
grid2 = {[-1 0 0.5 1], [10 15 20]};
G = pdbase(grid2, [1 1], 1);
physicalCellCount = G.ncell()
coefficientsPerCell = G.ncoeff()
physicalCells = G.cells()
```

```
physicalCellCount =
     6
coefficientsPerCell =
     4
physicalCells =
     1     1
     1     2
     2     1
     2     2
     3     1
     3     2
```



In ℓ dimensions, direction s has the one-based nodes $\mathcal{G}_s = \{\rho_s^{(1)}, \ldots, \rho_s^{(k_s)}\}$. The tensor grid is $\mathcal{G} = \mathcal{G}_1 \times \cdots \times \mathcal{G}_\ell$, with $\prod_{s=1}^{\ell} k_s$ physical nodes. A physical hypercube has the separate one-based multi-index $\mathbf{c} = (c_1, \ldots, c_\ell)$, so there are $\prod_{s=1}^{\ell}(k_s - 1)$ hypercubes. With the forward coordinates

$$\alpha_s = \frac{\rho_s - \rho_s^{(c_s)}}{\rho_s^{(c_s+1)} - \rho_s^{(c_s)}},$$

the local degree-m tensor representation is

$$C^{(\mathbf{c})}(\boldsymbol{\rho}) = \sum_{\mathbf{i} \in \{0,\ldots,m\}^\ell} \left(\prod_{s=1}^{\ell} B_{i_s}^m(\alpha_s) \right) C^{(\mathbf{c})}[\mathbf{i}].$$

The local Bernstein label $\mathbf{i}$ is zero-based, while every physical-node index and physical-cell component is one-based. Earlier dimensions vary more slowly in `cells`, `lbls`, `coeffs`, and rate-vertex traversal. Cell-local spline and gridding constructions for PD-LMIs motivate this architecture [3, 4]; Appendix A develops the underlying basis from degree-one interpolation to the general tensor case.

2

Installation, Verification, And Lookup

2.1 Installation

Start MATLAB in the repository root, or set `projectRoot` to the absolute path of the checked-out repository. Add the package recursively, then remove the documentation directory from the executable path.

Example

```
projectRoot = "path/to/Differential Parameter-Dependent Linear Matrix Inequality";
addpath(genpath(projectRoot));
rmpath(genpath(fullfile(projectRoot, "doc")));
```

For local work from the repository root, this package also resolves from:

Example

```
addpath(pwd)
```

2.2 Verification

Run the current MATLAB test entry point from the repository root:

Example

```
results = tests.run_all();
```

The test entry point covers helper utilities, `pdbase`, `pdmat`, `pdvar`, and ordinary `pdlmi` behavior. The YALMIP-backed `pdvar` and `pdlmi` tests require YALMIP. Solver smoke tests additionally require an SDP solver.

2.3 Smoke Test

First confirm that MATLAB resolves the four renamed constructors from this repository and not from another toolbox:

Example

```
which pdbase -all
which pdmat -all
which pdvar -all
which pdlmi -all
```

Then exercise known-data inspection and the solver-facing assembly boundary:

Example

```
A = pdmat({[0 1]}, {1, 3}, Degree=1);
T = bernsteinTable(A, "oneLine")
```

```
    CellSubscript      Expression
    -------------    ---------------

        {[1]}        "(1-alpha)*1 + alpha*3"
```

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric");
C = P >= 0;
F = toYalmip(C);
Cputinar = C.applyPutinar();
Fputinar = toYalmip(Cputinar);
Cbox = C.applyFullBoxPreorder();
Fbox = toYalmip(Cbox);
```

`F` is the direct coefficient certificate; `Fputinar` and `Fbox` are the default Putinar and full-box certificates for the same stored residual. Any one can be passed to ordinary YALMIP calls, for example:

Call forms

```
sol = optimize(F, objective, opts);
```

2.4 Global Function Lookup

Category	Entries
Backend	`pdbase`, constructor, `cells`, `lbls`, `coeffs`, `ncell`, `ncoeff`, `npar`, `size`
Known data	`pdmat`, construction, `evaluate`, `bernsteinTable`, `disp/display`, `plot`, operators, matrix methods
Decision variables	`pdvar`, construction, `rhodiff`, affine/mixed algebra, matrix methods, comparisons
LMIs	`pdlmi`, construction, direct coefficient assembly, `applyPolya`, `applyPutinar`, `applyFullBoxPreorder`, `toYalmip`, solver examples

3

pdbase Backend Reference

pdbase is the developer-facing backend parent for pdmat and pdvar. Ordinary users should model known data with pdmat and decision expressions with pdvar. This section documents pdbase because it explains the shared storage, traversal, and inspection contract used by both public modeling classes.

3.1 Lookup

Task	Function or field
Create backend object	pdbase
Inspect cells/labels	cells, lbls, coeffs
Count sizes	ncell, ncoeff, npar, size
Elevate stored coefficients	elevVals
Backend algebra mechanisms	elevLocalValues, bernElev, bernProd, mergeGrid
Read storage fields	GridInfo, LocalValues, metadata fields

3.2 pdbase Constructor

Syntax

Call forms

```
obj = pdbase(gridVectors, matrixSize, degree)
obj = pdbase(gridVectors, matrixSize, degree, localValues)
obj = pdbase(..., IsContinuous=tf)
obj = pdbase(..., ContainsDecision=tf)
obj = pdbase(..., HasRateDependence=tf)
obj = pdbase(..., RateBounds=rb)
obj = pdbase(..., SourceSummary=txt)
```

Arguments

- gridVectors is a cell vector with one strictly increasing node vector per parameter, such as {[0 1 2], [10 20]}.
- matrixSize is a positive two-element matrix size, such as [2 2].
- degree is a nonnegative integer Bernstein degree, such as 1.

- `localValues` is the nested physical-cell storage. When omitted, `pdbase` creates a zero coefficient-backed object.
- `IsContinuous`, `ContainsDecision`, and `HasRateDependence` are logical metadata scalars. They default to false; subclasses set them to describe their own payload contract.
- `RateBounds` is empty or an $\ell \times 2$ matrix of finite lower and upper rate bounds, such as `RateBounds=[-1 1; -2 2]`. Nonempty bounds also mark the object as rate-dependent.
- `SourceSummary` is inspectable origin text and defaults to `"coefficient-backed"`.

Examples and output

This backend example constructs a two-parameter parent object. The object has two physical cells because the first parameter has two intervals and the second parameter has one interval.

Example

```
obj = pdbase({[0 1 2], [10 20]}, [2 2], 1);
objectClass = class(obj)
matrixSize = obj.MatrixSize
degree = obj.Degree
cellCount = obj.ncell()
coefficientCount = obj.ncoeff()
parameterCount = obj.npar()
objectSize = size(obj)
```

```
objectClass =
    'pdbase'

matrixSize =
     2     2

degree =
     1

cellCount =
     2

coefficientCount =
     4

parameterCount =
     2

objectSize =
     2     2
```

The printed `ncoeff` value is $(1+1)^2 = 4$, one local Bernstein coefficient for each tensor-product label in the two parameter directions.

Remarks

- Malformed grids and node vectors raise `pdbase:InvalidGrid` or `pdbase:InvalidGridVector`.

- Invalid matrix sizes, degrees, or rate bounds raise `pdbase:InvalidMatrixSize`, `pdbase:InvalidDegree`, or `pdbase:InvalidRateBounds`.
- The nested tree must contain one leaf per physical cell and $(m+1)^\ell$ coefficient columns per leaf. Violations raise `pdbase:InvalidLocalValues` or `pdbase:InvalidCoefficientCell`.
- Payload matrices must have the declared size and be finite, real, and numeric or supported real YALMIP expressions. Malformed payloads raise `pdbase:InvalidCoefficientPayload`.
- Rate-affine payloads and `HasRateDependence=true` require compatible nonempty `RateBounds`.

3.3 Properties

Every `pdbase` property below is publicly readable through dot access but has private set access. Users inspect `obj.Degree` or `obj.GridInfo.Vectors`; construction and package algebra create new objects instead of assigning these properties.

Property	Meaning
`GridInfo`	Grid vectors, node counts, and grid bounds.
`MatrixSize`	Size of each matrix coefficient or matrix expression.
`Degree`	Local Bernstein degree stored in every physical cell.
`LocalValues`	Nested cell tree indexed by physical-cell subscript; each leaf stores coefficient cells.
`IsContinuous`	Metadata flag; subclasses are responsible for boundary sharing if continuity is true.
`ContainsDecision`	True when coefficient payloads include YALMIP decision variables.
`HasRateDependence`	True when the object carries rate metadata or rate-vertex rows.
`RateBounds`	Parameter-rate lower and upper bounds used by `rhodiff` and `pdlmi`.
`SourceSummary`	Short origin label such as `coefficient-backed`, `decision`, or `derivative`.

Example

```
obj = pdbase({[0 1 2], [10 20]}, [2 3], 1);
degree = obj.Degree
matrixSize = obj.MatrixSize
firstGrid = obj.GridInfo.Vectors{1}
source = obj.SourceSummary
```

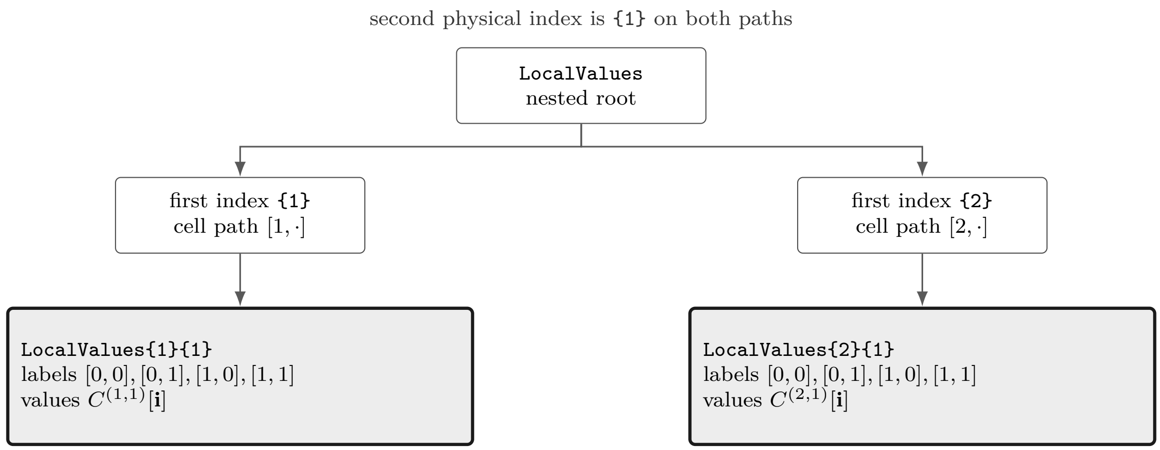


Figure 1: Nested cell-local storage for a two-parameter, degree-one object on `{[0 1 2], [10 20]}`. There are two physical cells, $[1, 1]$ and $[2, 1]$. Each leaf is a flat local coefficient cell in `lbls` order: $[0, 0]$, $[0, 1]$, $[1, 0]$, $[1, 1]$.

The nesting follows physical-cell subscripts, not the Bernstein labels. Start with `LocalValues\{i1\}`, then take one cell index per remaining parameter; this selects physical cell $[i_1, i_2, \ldots, i_\ell]$. Only after that selection does the leaf's flat cell array enumerate the $(m+1)^\ell$ local Bernstein coefficients. Thus `coeffs(obj, [2 1])` returns the lower leaf in fig. 1, while `lbls(obj)` explains the order inside that leaf.

3.4 Inspection: `cells`, `lbls`, And `coeffs`

Syntax

Call forms

```
subs = cells(obj)
subs = obj.cells()

labels = lbls(obj)
labels = obj.lbls()

c = coeffs(obj, cellSubs)
c = obj.coeffs(cellSubs)
```

Arguments

- `obj` is a `pdbase` object or a `pdmat`/`pdvar` subclass instance using the shared cell-local storage convention.
- `cellSubs` selects one physical cell for `coeffs`. Supply a numeric row vector such as `[2 1]` or a cell array of scalar subscripts such as `{2,1}`.
- `cells` returns physical hypercube subscripts in the shared traversal order; `lbls` returns local Bernstein labels in the shared coefficient order.
- `c` is the flat coefficient cell array for the selected physical cell, ordered consistently with `lbls(obj)`. Rate-dependent leaves return one row per active rate vertex in the stored row order.

Remarks

`coeffs` requires one positive integer physical-cell index per parameter, within the corresponding cell count. Invalid numeric or cell-array selectors raise `pdbase:InvalidCellSubs`.

Examples and output

The label order is shared by `pdmat`, `pdvar`, and `pdlmi`. In this two-parameter degree-one object, each physical cell stores four coefficient entries with labels $[0,0]$, $[0,1]$, $[1,0]$, and $[1,1]$.

Example

```
obj = pdbase({[0 1 2], [10 20]}, [1 1], 1);
labels = lbls(obj)
physicalCells = cells(obj)
c = coeffs(obj, [2 1]);
coefficientCount = numel(c)
```

```
labels =
     0     0
     0     1
     1     0
     1     1

physicalCells =
     1     1
     2     1

coefficientCount =
     4
```

3.5 Counts And Shape

Syntax

Call forms

```
n = ncell(obj)
n = obj.ncell()

n = ncoeff(obj)
n = obj.ncoeff()

n = npar(obj)
n = obj.npar()

sz = size(obj)
d = size(obj, dim)
```

Arguments

- `obj` is a `pdbase` object or subclass instance.
- `dim` is a positive integer matrix dimension for `size`; it follows MATLAB's ordinary `size(obj,dim)` convention.
- `ncell` returns $\prod_s (k_s - 1)$, the number of physical hypercube cells.
- `ncoeff` returns $(m+1)^\ell$, the number of local Bernstein coefficients stored per physical cell.
- `npar` returns the parameter dimension ℓ. `size` returns the matrix payload size, never the physical grid size.

Examples and output

For a two-parameter ($\ell = 2$) degree-one backend object, the first grid has two physical intervals and the second has one. The matrix payload is 2-by-3, independently of the grid shape. Thus $n_{\text{cell}} = (3-1)(2-1) = 2$ and each cell stores $n_{\text{coeff}} = (1+1)^2 = 4$ entries, ordered by the explicit labels $[0,0], [0,1], [1,0], [1,1]$.

Example

```
obj = pdbase({[0 1 2], [10 20]}, [2 3], 1);
physicalCellCount = obj.ncell()
coefficientCount = obj.ncoeff()
parameterCount = obj.npar()
matrixSize = size(obj)
```

```
rowCount = size(obj, 1)
columnCount = size(obj, 2)
thirdDimension = size(obj, 3)
```

```
physicalCellCount =
     2
coefficientCount =
     4
parameterCount =
     2
matrixSize =
     2     3
rowCount =
     2
columnCount =
     3
thirdDimension =
     1
```

3.6 Bernstein Storage Operations

The following entries separate whole-object elevation from the protected helpers used to align degrees, convolve coefficients, and refine grids. They change coefficient representations or temporary storage trees; they do not change the meanings of **`ncell`**, **`ncoeff`**, **`npar`**, or **`size`** described above.

3.6.1 `elevVals`

Syntax

Call forms

```
vals = obj.elevVals(degreeIncrement)
```

Arguments

- `degreeIncrement` is a finite nonnegative integer scalar. It is added to `obj.Degree` in every parameter direction; zero returns coefficient evidence at the current degree.

Description

`vals` is a nested `LocalValues`-shaped cell tree. Each physical cell contains coefficients of degree `obj.Degree + degreeIncrement`, with the represented polynomial, physical-cell tree, and any rate-row ordering preserved. The method does not modify `obj` or its stored coefficient evidence.

Remarks

An invalid increment raises `pdbase:InvalidDegreeIncrement`. A function-only `pdmat` has no stored Bernstein coefficient evidence, so this method raises `pdbase:MissingCoefficientEvidence`.

Relationship to backend and public algebra. Unlike protected `bernElev`, which elevates one flat cell coefficient family for internal alignment, `elevVals` is a public whole-object query returning elevated `LocalValues`. Public coefficient algebra uses `bernElev` internally when compatible operands need degree alignment; it does not mutate either operand.

3.6.2 `elevLocalValues`

Internal/protected mechanism. This method elevates a temporary nested coefficient tree after public algebra has remapped operands to a common physical grid. It is documented as backend context, not as a supported user call form.

Syntax

Call forms

```
vals = elevLocalValues(obj, vals, fromDeg, toDeg)
vals = elevLocalValues(obj, vals, fromDeg, toDeg, grid)
```

Each leaf must be a cell array with one column per degree-`fromDeg` Bernstein label. The method elevates each rate-vertex row independently to `toDeg`; it never mixes rows. The optional `grid` supplies the temporary physical-cell layout, and otherwise defaults to `obj.GridInfo.Vectors`. Equal source and target degrees return the input tree unchanged. A malformed leaf raises `pdbase:InvalidCoefficientCell`. Public users encounter this behavior through grid merging, assignment, concatenation, and arithmetic rather than by calling the protected method directly.

3.6.3 `bernElev`

Internal/protected mechanism. This signature documents the backend algorithm used by public algebra; it is not a supported user call form.

Syntax

Call forms

```
out = bernElev(obj, coeffs, fromDeg, toDeg)
```

Arguments

- `coeffs` is a flat coefficient-cell family in the same label order as `lbls(obj)`.
- `fromDeg` and `toDeg` are nonnegative integer degrees, with `toDeg >= fromDeg`.
- `out` contains $(\texttt{toDeg}+1)^{\ell}$ elevated coefficients in the same label order.

Examples and output

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 1]}, {1, 2, 3}, Degree=2);
C = A + B;
degree = C.Degree
```

```
degree =
     2
```

The example invokes the protected helper through public addition; users do not call `bernElev` directly. Invalid degree values raise `pdbase:InvalidDegree`; a target below the source degree raises `pdbase:InvalidDegreeElevation`; a coefficient count inconsistent with the source degree and parameter dimension raises `pdbase:InvalidCoefficientCell`.

3.6.4 `bernProd`

Internal/protected mechanism. This signature documents the backend algorithm used by public matrix multiplication; it is not a supported user call form.

Syntax

Call forms

```
out = bernProd(obj, lhs, lhsDeg, rhs, rhsDeg)
```

Arguments

- `lhs` and `rhs` are flat coefficient-cell families whose matrix payload inner dimensions are compatible for multiplication.
- `lhsDeg` and `rhsDeg` are their nonnegative local Bernstein degrees.
- `out` is the binomial-normalized coefficient convolution at degree `lhsDeg + rhsDeg`; payload products follow MATLAB matrix multiplication rules.

For one parameter, the protected helper implements

$$C[k] = \sum_{i+j=k} \frac{\binom{m}{i}\binom{n}{j}}{\binom{m+n}{k}} A[i]B[j].$$

It enumerates ordered payload products, groups them by the anti-diagonal $i + j = k$, and only then applies the binomial normalization. For tensor labels the same rule is applied componentwise. See the public `pdmat` multiplication entry in section 4.6.2, the `pdvar` known-data boundary in section 5.6, and the full derivation in Appendix A.

Examples and output

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
P = A * A;
degree = P.Degree
coefficients = cell2mat(P.coeffs(1))
```

```
degree =
     2
coefficients =
     1     2     4
```

The example invokes the protected helper through public matrix multiplication; users do not call `bernProd` directly. Even for scalar data, the middle coefficient comes from two ordered contributions rather than from one sample.

3.6.5 `mergeGrid`

Internal/protected mechanism. This signature documents the backend grid-refinement algorithm used by public algebra; it is not a supported user call form.

Syntax

Call forms

```
grid = obj.mergeGrid(errId, other1, other2, ...)
```

Arguments

- `errId` is the error identifier used when parameter counts or physical bounds are incompatible.
- `other1, other2, ...` are coefficient-backed operands, including function-backed operands constructed with an explicit `Degree`. They must have equal parameter counts and matching first and last nodes on every parameter axis.

- `grid` has the sorted union of the interior nodes. Public algebra re-expresses operands on these cells before degree elevation, coefficient-wise addition, or product convolution.

This protected helper is invoked by public algebra methods rather than called directly by users.

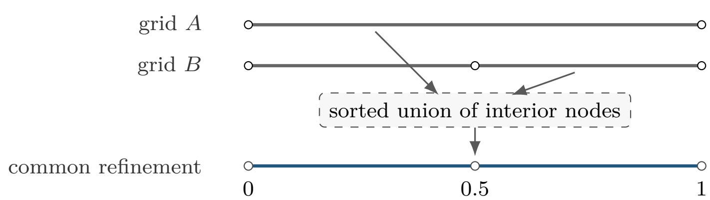
Examples and output

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 0.5 1]}, {10, 20, 30}, Degree=1);
C = A + B;
commonGrid = C.GridInfo.Vectors{1}
```

```
commonGrid =
         0    0.5000    1.0000
```

The example invokes the protected helper through public addition; users do not call `mergeGrid` directly.



No extra breakpoint is introduced: the sorted union is the unique minimal same-bound grid that contains every input node.

3.7 See Also

`pdmat`, `pdvar`, `cells`, `coeffs`, `lbls`, `ncell`, `ncoeff`, `npar`, `size`, `elevVals`, `elevLocalValues`, `bernElev`, `bernProd`, `mergeGrid`.

4

pdmat Reference

`pdmat` stores known finite real matrix data on a tensor grid. Objects can be coefficient-backed, function-only, or function-backed with explicit Bernstein coefficient evidence. Coefficient-backed objects participate in algebra. Function-only objects evaluate exactly through the stored handle, but do not expose coefficient evidence for `bernsteinTable` or coefficient algebra.

4.1 Lookup

Task	Entry
Construct data	`pdmat`
Evaluate	`evaluate`
Inspect/display	`bernsteinTable`, `disp`, `display`
Plot	`plot`
Arithmetic	`+`, `-`, unary `+/-`, `*`
Matrix/index methods	shape and structural methods, indexing and assignment

4.2 Construction

Syntax

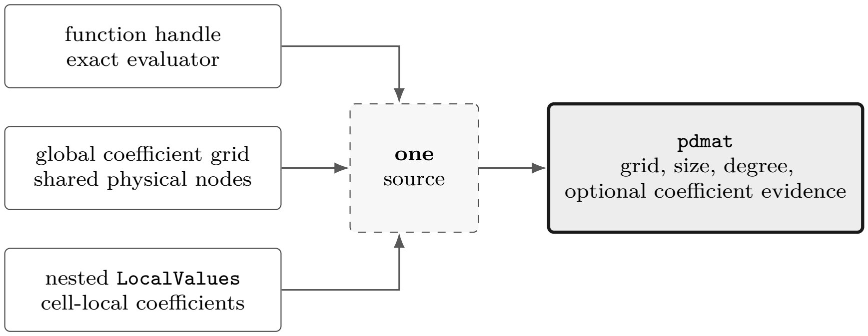
Call forms

```
A = pdmat(gridVector, source)
A = pdmat(gridVector, source, Degree=m)
A = pdmat(gridVectors, source)
A = pdmat(gridVectors, source, Degree=m)
```

Arguments

- `gridVector` is a numeric vector used as a one-parameter grid shorthand, such as `[0 1 2]`.
- `gridVectors` is a cell vector with one numeric grid vector per parameter, such as `{[0 1], [10 20]}`.
- `source` may be a function handle, a global cell grid of numeric Bernstein coefficients, or explicit nested `LocalValues`; for example, `{1, 2, 3}` is scalar degree-two data on one cell.

- `Degree=m` sets the local Bernstein degree. For function handles, omitting `Degree` creates a function-only object; passing a value such as `Degree=2` validates and stores Bernstein coefficient evidence while retaining the exact function handle for `evaluate`.



Examples and output

The scalar-grid shorthand is the shortest coefficient-backed form. The three numeric cells below are the global Bernstein coefficients for two physical cells with degree one.

Example

```
A = pdmat([0 1 2], {10, 20, 30}, Degree=1)
matrixSize = size(A)
degree = A.Degree
sourceSummary = A.SourceSummary
```

```
A =
  pdmat [1 1] over 1-D grid, degree 1, source coefficient-backed

matrixSize =
     1     1

degree =
     1

sourceSummary =
    "coefficient-backed"
```

For tensor grids, pass one grid vector per parameter. The source cell array matches the global coefficient grid implied by the two parameter directions.

Example

```
B = pdmat({[0 1], [10 20]}, {1 2; 3 4}, Degree=1)
secondGridVector = B.GridInfo.Vectors{2}
labels = B.lbls()
```

```
B =
  pdmat [1 1] over 2-D grid, degree 1, source coefficient-backed

secondGridVector =
    10    20

labels =
     0     0
     0     1
```

```
     1     0
     1     1
```

Use explicit nested `LocalValues` when each physical cell must be written directly. Here each cell stores two 1×2 row-vector coefficients.

Example

```
localVals = {{[1 0], [0 1]}, {[2 0], [0 2]}};
C = pdmat({[0 1 2]}, localVals, Degree=1)
matrixSize = size(C)
secondCellCoefficients = C.coeffs(2)
```

```
C =
  pdmat [1 2] over 1-D grid, degree 1, source coefficient-backed

matrixSize =
     1     2

secondCellCoefficients =
  1x2 cell array
    {[2 0]}    {[0 2]}
```

When `source` is a function handle and `Degree` is omitted, the object keeps the exact evaluator but does not claim coefficient evidence.

Example

```
F = pdmat({[0 1]}, @(rho) [rho rho^2])
sourceSummary = F.SourceSummary
evaluatedValue = F.evaluate(0.5)
```

```
F =
  pdmat [1 2] over 1-D grid, degree 1, source function

sourceSummary =
    "function"

evaluatedValue =
    0.5000    0.2500
```

Add `Degree` for a polynomial function when later coefficient inspection or algebra should use Bernstein evidence. The exact function handle is still used for point evaluation.

Example

```
G = pdmat({[0 1]}, @(rho) rho.^2, Degree=2)
sourceSummary = G.SourceSummary
firstCellCoefficients = G.coeffs(1)
```

```
G =
  pdmat [1 1] over 1-D grid, degree 2, source function-bernstein

sourceSummary =
    "function-bernstein"

firstCellCoefficients =
  1x3 cell array
    {[0]}    {[0]}    {[1]}
```

These forms all create finite known-data objects, but only the coefficient- backed and function-Bernstein forms can participate in coefficient algebra.

Description

FunctionHandle retains the supplied function handle for function-only and function-Bernstein sources; it is empty for cell-backed sources. Its set access is private.

Remarks

- Constructor option errors use pdmat:InvalidOptions, pdmat:UnsupportedOption, or pdmat:UnknownOption; grid errors use pdmat:InvalidGrid or pdmat:InvalidGridVector.
- Source and function validation uses pdmat:InvalidSource, pdmat:InvalidFunctionHandle, pdmat:InvalidFunctionOutput, or pdmat:InvalidData.
- Degree and storage failures use pdmat:InvalidDegree, pdmat:InvalidLocalValues, pdmat:InvalidCoefficientCell, or pdmat:InvalidCoefficientPayload.
- With explicit Degree, nonrepresentable functions raise pdmat:NonBernsteinPolynomial; invalid validation probes raise pdmat:InvalidFunctionValue.
- Mismatched shared faces are retained as discontinuous local data and emit warning pdmat:DiscontinuousLocalValues.

4.3 Properties

pdmat adds the read-only FunctionHandle property to the inherited pdbase properties in section 3.3. All use private set access. FunctionHandle contains the supplied handle for function and function-Bernstein sources and is empty for cell-backed sources. The inherited properties describe the grid, matrix size, Bernstein degree, cell-local evidence, continuity, and origin.

Example

```
A = pdmat({[0 1]}, @(rho) rho.^2, Degree=2);
degree = A.Degree
source = A.SourceSummary
hasFunction = ~isempty(A.FunctionHandle)
firstCell = A.LocalValues{1}
```

Assigning A.Degree, A.LocalValues, or A.FunctionHandle is not a supported update operation; construct a new pdmat instead.

4.4 `evaluate`

Syntax

Call forms

```
val = evaluate(A, point)
val = A.evaluate(point)
```

Arguments

- `A` is a `pdmat` object.
- `point` is a finite real scalar for a one-parameter object, such as `0.25`, or a finite real row vector with one entry per parameter, such as `[0.25 12]`. Every entry must lie within the corresponding grid bounds.
- `val` is a numeric matrix with `size(A)`. Coefficient-backed objects use local Bernstein interpolation; function-backed objects call their stored function handle.

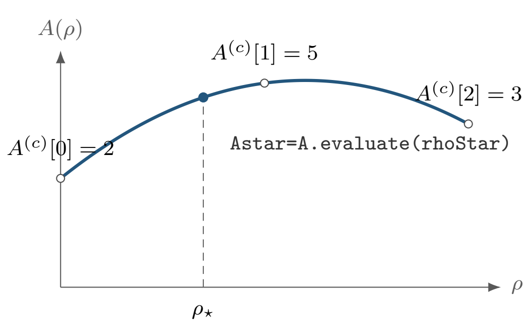
Remarks

A point with the wrong type, length, or finite-real shape raises `pdmat:InvalidPoint`; a point outside any grid bound raises `pdmat:PointOutOfBounds`. If a stored function fails or returns a nonfinite, complex, empty, or size-inconsistent matrix at the requested point, `evaluate` raises `pdmat:InvalidFunctionValue`.

For a degree-one scalar coefficient-backed object on $[\rho_1^{(1)}, \rho_1^{(2)}]$,

$$A(\rho) = (1-\alpha)A^{(c)}[0] + \alpha A^{(c)}[1], \qquad \alpha = \frac{\rho_1 - \rho_1^{(1)}}{\rho_1^{(2)} - \rho_1^{(1)}}.$$

The same local-coordinate rule is applied entrywise for matrix-valued data.



Evaluation follows the degree-two Bernstein curve with coefficients $[2, 5, 3]$*; the dashed line locates the requested physical point.*

Examples and output

Use the function-call form when the object is naturally one input among several MATLAB expressions.

Example

```
A = pdmat({[0 1]}, {2, 4}, Degree=1);
val = evaluate(A, 0.25)
```

```
val =
    2.5000
```

At $\rho = 0.25$, the public local coordinate is $\alpha = 0.25$, so the interpolated value is $0.75 \cdot 2 + 0.25 \cdot 4 = 2.5$. The dot-call form is equivalent and can be convenient in command-window exploration.

Example

```
A = pdmat({[0 1]}, {2, 4}, Degree=1);
val = A.evaluate(0.75)
```

```
val =
    3.5000
```

Function-backed objects route through the retained function handle, so non-polynomial expressions can still be evaluated exactly at a point.

Example

```
F = pdmat({[0 pi]}, @(rho) sin(rho));
val = F.evaluate(pi/2)
```

```
val =
     1
```

4.5 Visualization And Inspection

4.5.1 `bernsteinTable`

Syntax

Call forms

```
T = bernsteinTable(A)
T = bernsteinTable(A, cellSubs)
T = bernsteinTable(A, "oneLine")
T = bernsteinTable(A, cellSubs, "oneLine")
```

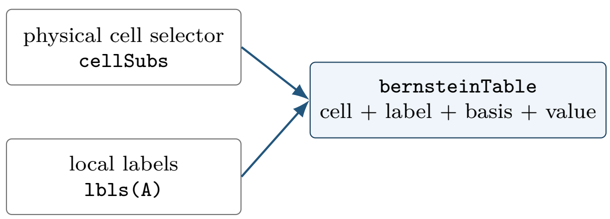
Arguments

- `cellSubs` selects one physical cell.
- `"oneLine"` returns one row per selected physical cell with a compact Bernstein expression.
- Without `"oneLine"`, the full table uses seven metadata columns; the output below shows their exact MATLAB names.

For one parameter, the `Basis` column reports

$$\binom{m}{j}(1-\alpha)^{m-j}\alpha^{j}.$$

For multiple parameters, the basis is the tensor product of that expression in each local coordinate, using `alpha1`, `alpha2`, and so on. `LocalIndex` is the local Bernstein label. `CoeffSubscript` maps the local label back to the global coefficient subscript on the physical grid. `IsPhysicalNode` is true when that coefficient lies on a physical grid node.



The table reports stored coefficient evidence. It does not sample a function and does not create solver constraints.

Examples and output

Example

```
A = pdmat({[0 1]}, {[0 1], [1 2]}, Degree=1);
T = bernsteinTable(A, 1, "oneLine")
```

CellSubscript	Expression
{[1]}	"(1-alpha)*[0 1] + alpha*[1 2]"

Use the full table when the metadata columns are the point of the example:

Example

```
A = pdmat({[0 1]}, {1, 2, 3}, Degree=2);
T = bernsteinTable(A)
```

TermIndex	CellSubscript	CoeffSubscript	LocalIndex	Basis	IsPhysicalNode	Value
1	{[1]}	{[1]}	{[0]}	"(1-alpha)^2"	true	{[1]}
2	{[1]}	{[2]}	{[1]}	"2(1-alpha)alpha"	false	{[2]}
3	{[1]}	{[3]}	{[2]}	"alpha^2"	true	{[3]}

Remarks

A function-only object has no coefficient evidence and raises `pdmat:FunctionOnlyBernsteinTable`. Unknown text options, repeated selectors, or invalid physical-cell selectors raise `pdmat:InvalidBernsteinTableInput`.

4.5.2 `disp` And `display`

Syntax

Call forms

```
disp(A)
display(A)
A
```

Arguments

`disp(A)` prints a compact one-line summary. `display(A)` is the command-window display used when an expression is not terminated by a semicolon; it prints grid, degree, coefficient, and source metadata.

Examples and output

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
disp(A)
display(A)
```

```
pdmat [1 1] over 1-D grid, degree 1, source coefficient-backed
A =
  pdmat with 1-by-1 matrix values
  Parameters: 1
    rho_1: [0, 1], 2 nodes
  Degree: 1
  Physical cells: 1
  Coefficients per cell: 2
  Continuous: true
  Source: coefficient-backed
  Function handle: false
```

4.5.3 `plot`

Syntax

Call forms

```
h = plot(A)
h = plot(A, dims)
h = plot(A, SamplesPerCell=n)
h = plot(A, dims, SamplesPerCell=n)
h = plot(A, ..., Name, Value)
```

Arguments

- `dims` is one or two parameter indices. For one-parameter objects, the default is `1`; otherwise the default is `[1 2]`, as in `plot(A, [1 2])`.
- `SamplesPerCell` is the package-owned sampling option. The default is `15`; use `SamplesPerCell=4` for a coarse diagnostic plot.
- Remaining name-value pairs pass through to MATLAB graphics. One- dimensional plots pass them to `plot`; two-dimensional plots pass them to `surf`. Examples include `LineWidth`, `FaceAlpha`, and `EdgeColor`.
- The output `h` is the vector of graphics handles, one entry per matrix element.

Remarks

An invalid parameter index, a repeated dimension, or a request for more than two plotting dimensions raises `pdmat:InvalidPlotDimensions`. Malformed plotting options or a nonpositive integer `SamplesPerCell` raise `pdmat:InvalidPlotOptions`. Errors from MATLAB graphics options remain MATLAB graphics errors. Every unselected parameter is fixed at its lower grid bound, `A.GridInfo.Bounds(:,1)`. A plot of a three-parameter object is therefore a one- or two-dimensional slice, not a volume plot.

Examples and output

Each source listing below is the complete source used by `generate_plot_figures.m` for the adjacent result.

One-dimensional entries.

Example

```
A = pdmat({[-1 1]}, @(rho) [rho, rho.^2]);
plot(A, SamplesPerCell=80, LineWidth=2)
grid on
```

```
xlabel("rho")
ylabel("matrix entry")
title("One-dimensional pdmat entries")
legend(["A(1,1)", "A(1,2)"], "Location", "eastoutside")
```

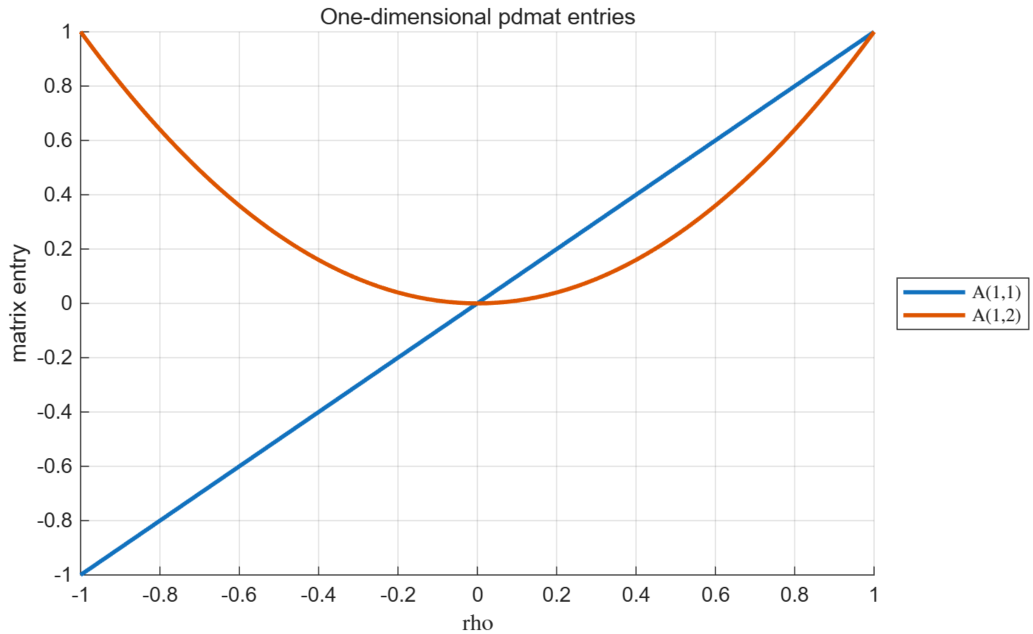


Figure 2: Result of the one-dimensional source above. The method returns one line handle per matrix entry.

Two-dimensional surface.

Example

```
A = pdmat({[-1 1], [0 2]}, ...
    @(rho1, rho2) 1 + rho1.^2 + 0.5*rho2 + rho1.*rho2);
plot(A, [1 2], SamplesPerCell=45, EdgeColor="none")
grid on
view(35, 25)
xlabel("rho1", "Interpreter", "none")
ylabel("rho2", "Interpreter", "none")
zlabel("matrix entry")
title("Two-dimensional pdmat surface")
```

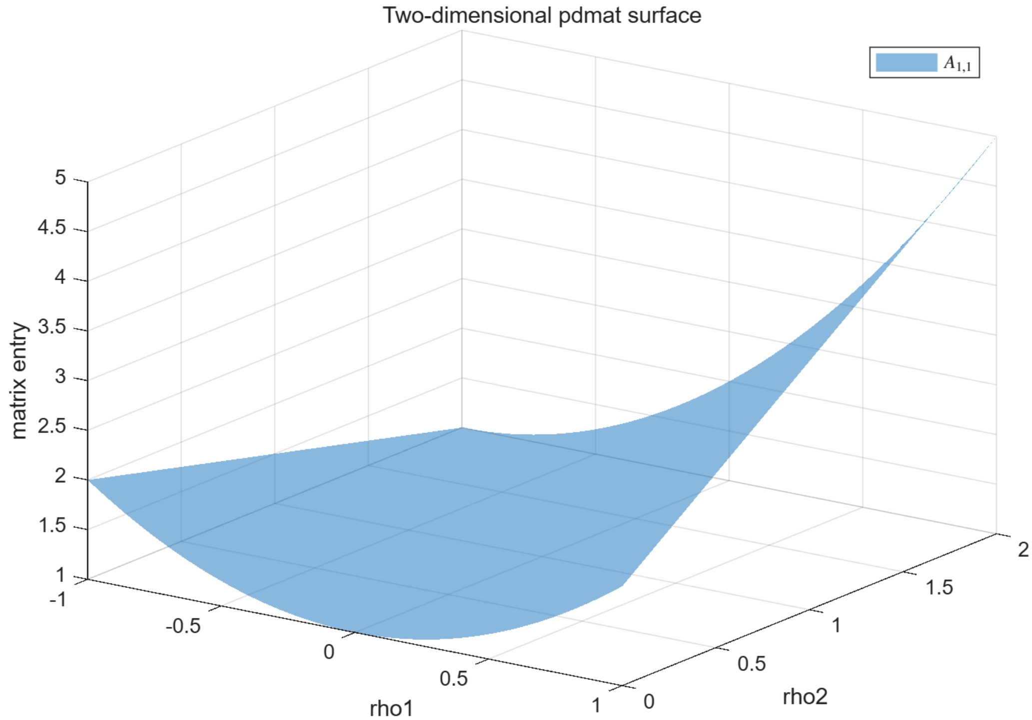


Figure 3: Result of the two-dimensional source above.

Two-dimensional slice of a three-parameter object.

Example

```
A = pdmat({[−1 1], [0 2], [−2 3]}, ...
    @(rho1, rho2, rho3) rho1.^2 + rho2 + 2∗rho3);
plot(A, [1 2], SamplesPerCell=45, EdgeColor="none")
grid on
view(35, 25)
title("Three−parameter object: rho3 fixed at −2")
```

Here $\rho_3 = -2$, the lower bound in the third row of `A.GridInfo.Bounds`. The plotted surface is $\rho_1^2 + \rho_2 - 4$.

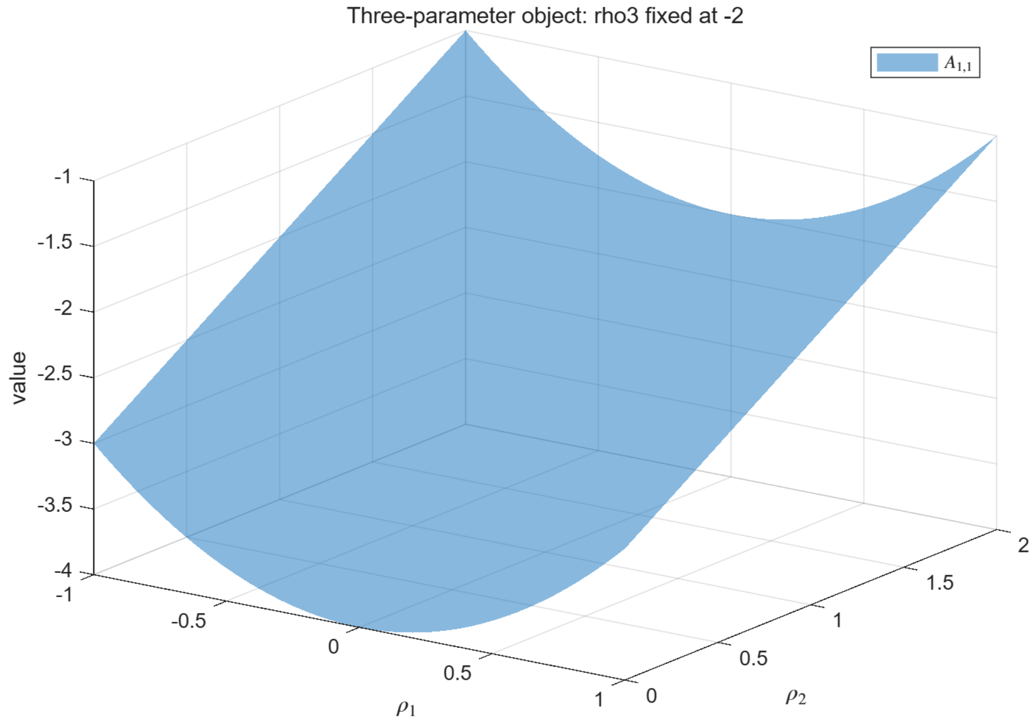


Figure 4: Result of the three-parameter slice source above; the plotting API still selects only dimensions 1 and 2.

4.6 Algebra And Matrix Operations

A `pdmat` operation acts on complete matrix coefficients while the physical grid remains attached to the result. The blocks below deliberately show one operation family at a time: purpose, supported syntax, input, immediate output, and—when the result is spatial—a before/after picture.

4.6.1 Signs, Addition, And Subtraction

Use signs, sums, and differences to build known matrix expressions. Compatible physical bounds are refined to the union grid, and lower-degree operands are elevated exactly before matching coefficients are combined. The corresponding MATLAB overload names are `plus`, `minus`, `uplus`, and `uminus`.

Syntax

Call forms

```
S = A + B
S = A + M
S = M + A
D = A - B
D = A - M
D = M - A
P = +A
N = -A
```

Here M is a finite real scalar or size-compatible matrix. A scalar broadcasts to the matrix payload size. The first example makes the degree alignment visible.

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 1]}, {10, 20, 30}, Degree=2);
S = A + B;
sumDegree = S.Degree
sumCoefficients = cell2mat(S.coeffs(1))
```

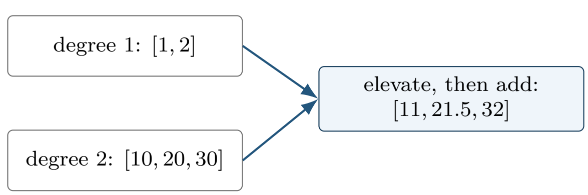
```
sumDegree =
     2
sumCoefficients =
   11.0000   21.5000   32.0000
```

The middle coefficient of the elevated A is $(1+2)/2 = 1.5$, so the middle sum is $1.5 + 20 = 21.5$. Subtraction and unary signs act immediately on the same local coefficients.

Example

```
D = 5 - A;
N = -A;
differenceCoefficients = cell2mat(D.coeffs(1))
negativeAtOne = N.evaluate(1)
positiveClass = class(+A)
```

```
differenceCoefficients =
     4     3
negativeAtOne =
    -2
positiveClass =
    'pdmat'
```



When grids have the same outer bounds but different interior nodes, the union grid is used before addition or subtraction. For example, `pdmat({[0 1]},...)+pdmat({[0 .5 1]},...)` has grid `[0 .5 1]`; each operand is represented exactly on both resulting cells.

4.6.2 Ordered Matrix Multiplication

Use `mtimes` for a matrix product. The inner payload dimensions must match. Two parameter-dependent factors produce the degree sum; a numeric factor is constant and does not raise the degree.

Syntax

Call forms

```
K = A * B
K = A * M
K = M * A
```

For one parameter, fix physical cell c and let $A = \sum_i B_i^m A^{(c)}[i]$ and $D = \sum_j B_j^n D^{(c)}[j]$. Then $K = AD$ has coefficients

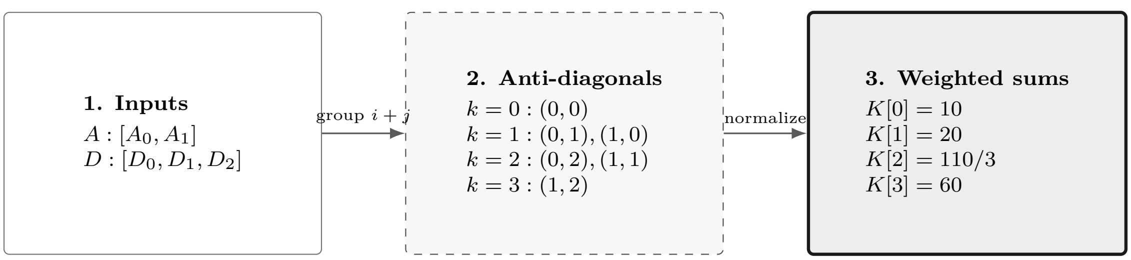
$$K^{(c)}[k] = \sum_{i+j=k} \frac{\binom{m}{i}\binom{n}{j}}{\binom{m+n}{k}} A^{(c)}[i] D^{(c)}[j].$$

The left/right matrix order is never exchanged.

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 1]}, {10, 20, 30}, Degree=2);
K = A * B;
productDegree = K.Degree
productCoefficients = cell2mat(K.coeffs(1))
```

```
productDegree =
     3
productCoefficients =
   10.0000   20.0000   36.6667   60.0000
```



Coefficient labels add under the weighted convolution; payload matrices multiply in the written order. The three panels make the convolution readable without crossing contribution arrows.

For tensor data, label addition and binomial weights are applied independently in each parameter direction. If $\mathbf{m}, \mathbf{n} \in \mathbb{N}^\ell$ are direction-wise basis degrees, the ℓ-dimensional ordered convolution is

$$K^{(\mathbf{c})}[\mathbf{k}] = \sum_{\mathbf{i}+\mathbf{j}=\mathbf{k}} \left(\prod_{s=1}^{\ell} \frac{\binom{m_s}{i_s}\binom{n_s}{j_s}}{\binom{m_s+n_s}{k_s}} \right) A^{(\mathbf{c})}[\mathbf{i}] D^{(\mathbf{c})}[\mathbf{j}].$$

Two degree-one factors in two parameters produce degree two and $3^2 = 9$ coefficients per physical cell.

Remarks

Incompatible operands for `+`, `-`, and `*` raise `pdmat:InvalidAddition`, `pdmat:InvalidSubtraction`, or `pdmat:InvalidMultiplication`. Parameter counts or outer bounds that cannot be merged raise `pdmat:MixedGrid`. Coefficient algebra on a function-only object raises `pdmat:FunctionOnlyAlgebra`; construct the function with an explicit verified `Degree` when coefficient evidence is required.

4.7 Matrix Methods

These methods transform every local matrix coefficient in the same way. They do not transpose, reshape, or concatenate the physical grid. Except for the documented evaluator/display/shape paths, structural methods require coefficient evidence.

4.7.1 Shape, Equality, And Display Protocols

Shape queries describe the matrix payload, not the grid. Equality compares coefficient-backed functions after compatible grid refinement and degree elevation. The protocol methods support ordinary MATLAB indexing syntax and dot access; they are not a second storage API.

Syntax

Call forms

```
sz = size(A)
d = size(A, dim)
n = height(A)
n = width(A)
n = length(A)
n = numel(A)
n = ndims(A)
tf = isequal(A, B, ...)
idx = end(A, k, n)
n = numArgumentsFromSubscript(A, S, ctx)
```

Example

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
matrixSize = size(A)
rowCount = height(A)
columnCount = width(A)
longestDimension = length(A)
elementCount = numel(A)
dimensionCount = ndims(A)
equalsUnaryPlus = isequal(A,+A)
```

```
matrixSize =
     2     3
rowCount =
```

```
     2
columnCount =
     3
longestDimension =
     3
elementCount =
     6
dimensionCount =
     2
equalsUnaryPlus =
  logical
   1
```

A function-only object can use shape queries and compares its stored metadata and function handle. A coefficient-backed equality that cannot form a common comparison grid raises `pdmat:InvalidEquality`. The detailed multi-line `display` transcript is shown in section 4.5.2.

4.7.2 Transpose And Conjugate Transpose

Both transpose forms act on every coefficient. Because `pdmat` payloads are finite and real, conjugation does not change their values.

Syntax

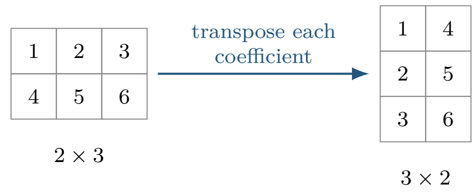
Call forms

```
T = transpose(A)
H = ctranspose(A)
T = A.'
H = A'
```

Example

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
T = A.';
H = A';
transposeSize = size(T)
formsEqual = isequal(T,H)
transposeAtZero = T.evaluate(0)
```

```
transposeSize =
     3     2
formsEqual =
  logical
   1
transposeAtZero =
     1     4
     2     5
     3     6
```



4.7.3 Vectorization, Reshape, And Squeeze

Use these methods to change the matrix view while preserving column-major element order, the parameter grid, the degree, and every coefficient value. `reshape` accepts exactly two positive dimensions with at most one inferred `[]` dimension. `squeeze` retains the package's two-dimensional matrix convention.

Syntax

Call forms

```
V = vec(A)
R = reshape(A, [m n])
R = reshape(A, m, n)
S = squeeze(A)
```

Example

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
V = vec(A);
R = reshape(A, [3 2]);
S = squeeze(A);
vectorSize = size(V)
reshapeSize = size(R)
squeezeSize = size(S)
vectorAtZero = V.evaluate(0).'
```

```
vectorSize =
     6     1
reshapeSize =
     3     2
squeezeSize =
     2     3
vectorAtZero =
     1     4     2     5     3     6
```

Malformed or element-count-changing reshape requests raise `pdmat:InvalidReshape`.

4.7.4 Diagonals And Trace

`diag` extracts a selected diagonal from a matrix, or builds a square diagonal matrix from a vector. A finite integer offset `k` may not select an empty diagonal. `trace` returns a 1-by-1 `pdmat`.

Syntax

Call forms

```
d = diag(A)
d = diag(A, k)
D = diag(v)
D = diag(v, k)
t = trace(A)
```

Example

```
A = pdmat({[0 1]}, {[1 2; 3 4], [10 20; 30 40]}, Degree=1);
d = diag(A);
D = diag(d);
t = trace(A);
diagonalAtZero = d.evaluate(0)
rebuiltAtZero = D.evaluate(0)
traceAtZero = t.evaluate(0)
```

```
diagonalAtZero =
     1
     4
rebuiltAtZero =
     1     0
     0     4
traceAtZero =
     5
```

Invalid offsets or empty selections raise `pdmat:InvalidDiag`.

4.7.5 Triangular Parts And Reductions

Triangular extraction preserves or zeros entries relative to a selected diagonal. Reductions act along matrix dimensions; `"all"` is supported by `sum` and `mean`.

Syntax

Call forms

```
L = tril(A)
L = tril(A, k)
U = triu(A)
U = triu(A, k)
S = sum(A)
S = sum(A, dim)
S = sum(A, "all")
M = mean(A)
M = mean(A, dim)
M = mean(A, "all")
C = cumsum(A)
C = cumsum(A, dim)
```

First inspect the two triangular results.

Example

```
A = pdmat({[0 1]}, {[1 2; 3 4], [10 20; 30 40]}, Degree=1);
L = tril(A);
U = triu(A);
lowerAtZero = L.evaluate(0)
upperAtZero = U.evaluate(0)
```

```
lowerAtZero =
     1     0
     3     4
upperAtZero =
     1     2
     0     4
```

Then apply reductions to the same coefficient matrix.

Example

```
rowSum = sum(A, 2);
allMean = mean(A, "all");
running = cumsum(A, 2);
rowSumAtZero = rowSum.evaluate(0)
allMeanAtZero = allMean.evaluate(0)
runningAtZero = running.evaluate(0)
```

```
rowSumAtZero =
     3
     7
allMeanAtZero =
    2.5000
runningAtZero =
     1     3
     3     7
```

Invalid triangular offsets raise `pdmat:InvalidTriangularPart`. Invalid reduction calls raise `pdmat:InvalidSum`, `pdmat:InvalidMean`, or `pdmat:InvalidCumsum`.

4.7.6 Reordering And Rotation

These methods reorder rows or columns inside every coefficient; they never reverse the scheduling grid.

Syntax

Call forms

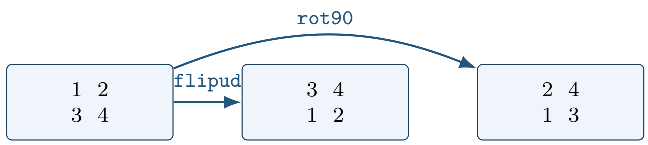
```
B = flip(A)
B = flip(A, dim)
B = flipud(A)
B = fliplr(A)
B = rot90(A)
```

```
B = rot90(A, k)
```

Example

```
A = pdmat({[0 1]}, {[1 2; 3 4], [10 20; 30 40]}, Degree=1);
F = flipud(A);
R = rot90(A);
flippedAtZero = F.evaluate(0)
rotatedAtZero = R.evaluate(0)
```

```
flippedAtZero =
     3     4
     1     2
rotatedAtZero =
     2     4
     1     3
```



Invalid dimensions or quarter-turn counts raise `pdmat:InvalidFlip` or `pdmat:InvalidRot90`.

4.7.7 Concatenation And Block Assembly

Concatenation joins payload matrices after compatible grid refinement and degree elevation. Numeric blocks are promoted to constant known data. `cat` supports only dimensions 1 and 2.

Syntax

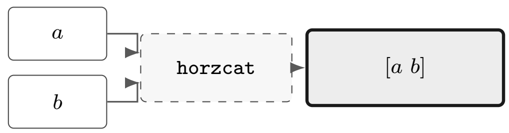
Call forms

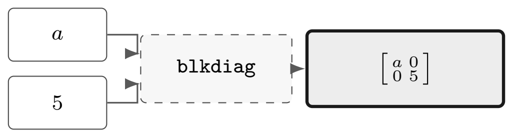
```
H = horzcat(A, B, ...)
V = vertcat(A, B, ...)
H = [A, B]
V = [A; B]
C = cat(dim, A, B, ...)
D = blkdiag(A, B, ...)
```

Example

```
a = pdmat({[0 1]}, {1, 2}, Degree=1);
b = pdmat({[0 1]}, {10, 20}, Degree=1);
H = [a, b];
V = [a; b];
D = blkdiag(a, 5);
horizontalAtZero = H.evaluate(0)
verticalAtZero = V.evaluate(0)
blockDiagonalAtZero = D.evaluate(0)
```

```
horizontalAtZero =
     1    10
verticalAtZero =
     1
    10
blockDiagonalAtZero =
     1     0
     0     5
```





Incompatible blocks or syntax raise `pdmat:InvalidConcatenation`; unsupported dimensions raise `pdmat:UnsupportedCatDimension`; invalid block-diagonal inputs raise `pdmat:InvalidBlkdiag`. Function-only operands raise `pdmat:FunctionOnlyAlgebra`.

4.7.8 Repetition

`repmat` tiles the matrix payload. It accepts a positive scalar `m`, interpreted as `[m m]`, two positive counts, or a two-element count vector. Grid and Bernstein degree remain unchanged.

Syntax

Call forms

```
R = repmat(A, m)
R = repmat(A, m, n)
R = repmat(A, [m n])
```

Example

```
v = pdmat({[0 1]}, {[1 2], [3 4]}, Degree=1);
R = repmat(v, [2 2]);
repeatedSize = size(R)
repeatedDegree = R.Degree
repeatedAtZero = R.evaluate(0)
```

```
repeatedSize =
     2     4
repeatedDegree =
     1
repeatedAtZero =
     1     2     1     2
     1     2     1     2
```

Invalid repetition shapes raise `pdmat:InvalidRepmat`.

4.8 Indexing And Assignment

Parenthesis indexing selects a matrix block from every coefficient. Assignment writes the selected block into every coefficient after compatible promotion and degree alignment. It does not select a physical cell; use `coeffs` for cell inspection.

Syntax

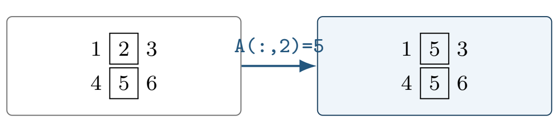
Call forms

```
B = A(rows, cols)
B = A(1:end, end)
value = A.PropertyName
A(rows, cols) = numericValue
A(rows, cols) = B
```

Example

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
lastCol = A(:, end);
A(:, 2) = 5;
selectedSize = size(lastCol)
assignedSize = size(A)
selectedAtZero = lastCol.evaluate(0)
assignedAtZero = A.evaluate(0)
```

```
selectedSize =
     2     1
assignedSize =
     2     3
selectedAtZero =
     3
     6
assignedAtZero =
     1     5     3
     4     5     6
```



`subsref`, `subsasgn`, `end`, and `numArgumentsFromSubscript` implement this two-index MATLAB protocol. Invalid selectors raise `pdmat:InvalidSubscript`. Linear indexing, deletion, or non-parenthesis assignment raises `pdmat:UnsupportedAssignment`; an incompatible right-hand side raises `pdmat:InvalidAssignment`. Slicing and assignment require coefficient evidence and otherwise raise `pdmat:FunctionOnlyAlgebra`.

4.9 Remarks

1. Function-only objects created without `Degree` do not have Bernstein coefficient evidence for `bernsteinTable` or coefficient algebra.

2. Numeric operands must be finite real nonempty matrices with compatible sizes, or finite real scalars that can broadcast to the required size.

3. Grid refinement is supported for matching parameter dimensions and matching grid bounds.

4.9.1 See Also

`pdmat`, `evaluate`, `bernsteinTable`, `plot`, `pdbase`.

5

pdvar Reference

`pdvar` creates continuous cell-local Bernstein decision expressions backed by YALMIP `sdpvar` coefficients. It participates in supported affine and known-data algebra. It does not choose solvers or call `optimize`; solver-facing work happens after comparison overloads create `pdlmi` objects and `toYalmip` converts them to YALMIP constraints.

5.1 Lookup

Task	Entry
Construct decisions	`pdvar`
Inspect coefficients	`bernsteinTable`
Convert assigned coefficients	`value`
Differentiate rates	`rhodiff`
Affine/mixed algebra	`+`, `-`, `*` with known data
Matrix/index methods	structural methods, indexing and assignment
Build LMIs	`<=`, `>=`

5.2 Construction

Syntax

Call forms

```
P = pdvar(n, gridVectors)
P = pdvar(n, n, gridVectors)
P = pdvar(n, m, gridVectors)
P = pdvar(..., "full")
P = pdvar(..., "symmetric")
P = pdvar(..., Degree=0)
P = pdvar(..., Degree=1)
P = pdvar(..., Degree=d)
P = pdvar(..., RateBounds=rb)
```

Arguments

- `pdvar(n, gridVectors)` creates an $n \times n$ square variable using scalar-size shorthand; for example, `pdvar(2, {[0 1]})`. Square variables default to symmetric YALMIP coefficients.

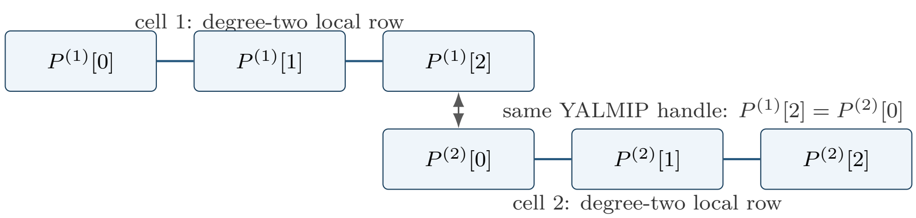
- `pdvar(n, n, gridVectors)` is the explicit square-size form; for example, `pdvar(2, 2, {[0 1]})`. It has the same default symmetric structure as the shorthand.

- `pdvar(n, m, gridVectors)` creates an $n \times m$ variable. Rectangular variables default to full YALMIP coefficients; for example, `pdvar(2, 3, {[0 1]})`.

- `"full"` and `"symmetric"` explicitly choose the YALMIP coefficient structure. `"symmetric"` requires a square size.

- `Degree=1` is the default continuous Bernstein decision variable. Every finite nonnegative integer `Degree=d` is supported. `Degree=0` creates one parameter-independent decision matrix shared across the grid; `d >= 1` creates continuous piecewise Bernstein data whose neighboring cells share complete control faces.

- `RateBounds=rb` stores parameter-rate metadata for later `rhodiff(P)` calls. Use `RateBounds=[-1 1]` for one parameter; use a two-row matrix for two parameters, as shown in the rate-metadata example below.



For positive degree, neighboring physical cells reuse complete boundary-face YALMIP handles; no extra continuity LMI is generated.

Examples and output

The square shorthand creates a 2×2 continuous decision object. The default degree is one and the object carries no rate metadata.

Example

```
yalmip('clear')
Ps = pdvar(2, {[0 1]});
variableClass = class(Ps)
matrixSize = size(Ps)
degree = Ps.Degree
rateDependent = Ps.HasRateDependence
```

```
variableClass =
    'pdvar'

matrixSize =
     2     2

degree =
     1

rateDependent =
  logical
   0
```

Use two dimension arguments for a rectangular full variable.

Example

```
yalmip('clear')
Pf = pdvar(2, 3, {[0 1]});
variableClass = class(Pf)
matrixSize = size(Pf)
degree = Pf.Degree
```

```
variableClass =
    'pdvar'

matrixSize =
     2     3

degree =
     1
```

The structure flag makes the coefficient structure explicit. This is useful when a square variable should use full, not symmetric, coefficients.

Example

```
yalmip('clear')
Pfull = pdvar(2, 2, {[0 1]}, "full");
variableClass = class(Pfull)
matrixSize = Pfull.MatrixSize
degree = Pfull.Degree
```

```
variableClass =
    'pdvar'

matrixSize =
     2     2

degree =
     1
```

Use `Degree=0` for a parameter-independent decision variable stored with grid metadata.

Example

```
yalmip('clear')
P0 = pdvar(1, [0 1 2], Degree=0);
variableClass = class(P0)
degree = P0.Degree
cellCount = P0.ncell()
```

```
variableClass =
    'pdvar'

degree =
     0

cellCount =
     2
```

Rate bounds are metadata for later `rhodiff(P)` calls. `pdlmi` constrains rate vertices only after an expression such as `rhodiff` has stored rate-vertex rows.

Example

```
yalmip('clear')
Pr = pdvar(1, {[0 1], [10 20]}, RateBounds=[−1 2; −3 4]);
variableClass = class(Pr)
rateDependent = Pr.HasRateDependence
rateBounds = Pr.RateBounds
```

```
variableClass =
    'pdvar'

rateDependent =
  logical
   1

rateBounds =
    -1     2
    -3     4
```

Remarks

- Missing or misplaced dimensions and grids raise `pdvar:InvalidInput`; invalid sizes and degrees raise `pdvar:InvalidMatrixSize` or `pdvar:InvalidDegree`.
- Option failures use `pdvar:InvalidOptions`, `pdvar:UnknownOption`, or `pdvar:Unsupported Option`.
- A symmetric nonsquare request raises `pdvar:InvalidStructure`; grid failures use `pdvar: InvalidGrid` or `pdvar:InvalidGridVector`.
- `RateBounds` must be a finite ℓ-by-2 matrix with each lower bound no greater than its upper bound; violations raise `pdbase:InvalidRateBounds`.

5.3 Properties

`pdvar` declares no additional class-owned properties. It exposes the read-only inherited `pdbase` properties listed in section 3.3; their private set access prevents user assignment. In particular, decision coefficients are inspected through `P.LocalValues`, while `P.Degree`, `P.IsContinuous`, and `P.RateBounds` describe their representation and rate metadata.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric", Degree=2, RateBounds=[−1 1]);
degree = P.Degree
isContinuous = P.IsContinuous
rateBounds = P.RateBounds
firstCellCoefficients = P.LocalValues{1}
```

Use public algebra to create a modified value; direct assignments such as `P.Degree = 1` are not supported.

5.4 `rhodiff`

Syntax

Call forms

```
D = rhodiff(P)
D = rhodiff(P, rb)
```

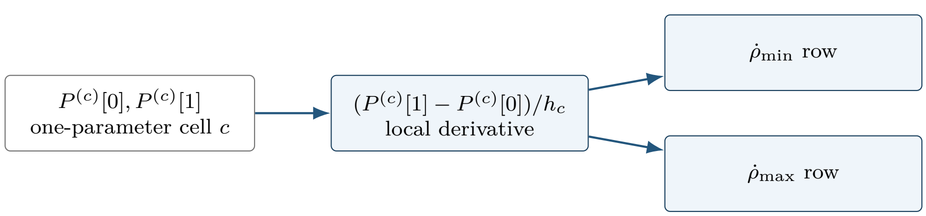
Arguments

- `P` is a `pdvar` without existing rate-vertex rows.
- `rb` is a finite ℓ-by-2 numeric matrix of lower and upper parameter-rate bounds, with each lower bound no greater than its upper bound. It must equal `P.RateBounds` when the object already carries nonempty bounds.
- Without `rb`, `rhodiff(P)` requires nonempty `P.RateBounds`.
- `D` is a discontinuous `pdvar` with one coefficient row per rate vertex. In one parameter a degree-m derivative has degree $m-1$, while a degree-zero derivative remains zero; multivariate partials are elevated into the implementation's common tensor degree.

For a one-parameter degree-one coefficient pair $P^{(c)}[0], P^{(c)}[1]$ on physical cell c of width h_c, the rate-dependent derivative row is

$$\dot{P}(\rho, \dot{\rho}) = \dot{\rho}\,\frac{P^{(c)}[1] - P^{(c)}[0]}{h_c}.$$

The implementation stores one row for each allowed rate vertex.



Differentiation changes local coefficients; rate substitution creates rows. Neither operation adds physical cells.

Remarks

Unsupported call arity or differentiating an expression that already contains rate-vertex rows raises `pdvar:InvalidDiff`. Explicit malformed rate bounds raise `pdvar:InvalidRateBounds`. Calling `rhodiff(P)` without stored bounds raises `pdvar:MissingRateBounds`; explicit bounds that disagree with `P.RateBounds` raise `pdvar:RateBoundsMismatch`.

Examples and output

When rate bounds live on the object, `rhodiff(P)` uses them directly.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1 2]}, "symmetric", RateBounds=[-1 1]);
D = rhodiff(P);
derivativeClass = class(D)
derivativeDegree = D.Degree
isContinuous = D.IsContinuous
coefficientTableSize = size(D.coeffs(1))
rateBounds = D.RateBounds
```

```
derivativeClass =
    'pdvar'
derivativeDegree =
     0
isContinuous =
  logical
   0
coefficientTableSize =
     2     1
rateBounds =
    -1     1
```

For a two-parameter object, the rate-vertex rows enumerate all lower/upper combinations from the two rows of `RateBounds`.

Example

```
yalmip('clear')
Q = pdvar(1, {[0 1], [10 20]}, RateBounds=[-1 1; -2 2]);
E = rhodiff(Q);
derivativeClass = class(E)
derivativeDegree = E.Degree
coefficientTableSize = size(E.coeffs([1 1]))
rateBounds = E.RateBounds
```

```
derivativeClass =
    'pdvar'
derivativeDegree =
     1
coefficientTableSize =
     4     4
rateBounds =
    -1     1
    -2     2
```

5.5 `bernsteinTable`

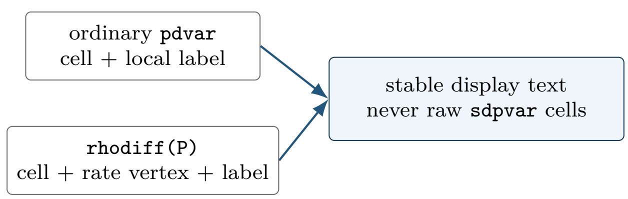
Syntax

Call forms

```
T = bernsteinTable(P)
T = bernsteinTable(P, cellSubs)
T = bernsteinTable(P, "oneLine")
T = bernsteinTable(P, cellSubs, "oneLine")
```

Arguments

- `P` is a `pdvar` object.
- `cellSubs` selects one physical cell; omit it to list every cell.
- `"oneLine"` is the only supported text option. It returns one compact Bernstein expression per selected cell.
- `T` is a table. Ordinary rows contain the local label, basis, physical-node flag, and symbolic or numeric value. A rate-dependent derivative also includes `RateVertexIndex` and `RateVertex`.



Examples and output

Example

```
yalmip('clear')
P = pdvar(1, {[0 1]});
T = bernsteinTable(P, "oneLine")
```

```
    CellSubscript       Expression
    -------------    ----------------

        {[1]}        "(1-alpha)*internal(1) + alpha*internal(2)"
```

Remarks

`bernsteinTable` is an inspection method, not a solver call. Text options other than `"oneLine"`, repeated selectors, invalid physical-cell selectors, or inconsistent rate-vertex rows raise `pdvar:InvalidBernsteinTableInput`.

5.6 Affine Algebra And Known-Data Products

A `pdvar` result must remain affine in the underlying YALMIP decisions. The following blocks separate affine sums from products so the supported boundary is visible beside each example.

5.6.1 Signs, Addition, And Subtraction

Use this family to combine compatible decision expressions, coefficient-backed known data, finite real numeric data, and affine `sdpvar` matrices. Operands with the same physical bounds are aligned on a common grid and degree. The corresponding overload names are `plus`, `minus`, `uplus`, and `uminus`.

Syntax

Call forms

```
R = P + Q
R = P + A
R = A + P
R = P + M
R = M + P
R = P + X
R = X + P
R = P - Q
R = P - A
R = A - P
R = P - M
R = M - P
R = +P
R = -P
```

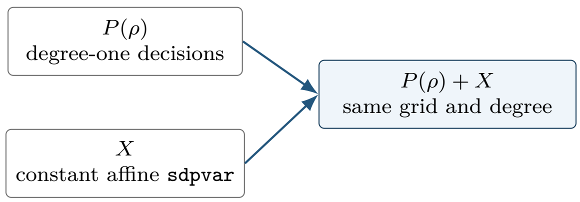
Here `A` is coefficient-backed `pdmat`, `M` is numeric, and `X` is affine `sdpvar`. A numeric scalar broadcasts to the payload size; an affine `sdpvar` is promoted as parameter-independent data.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "full");
X = sdpvar(2, 2, 'full');
S = P + X;
D = eye(2) — P;
N = —P;
sumClass = class(S)
sumSize = size(S)
sumDegree = S.Degree
sumContainsDecision = S.ContainsDecision
differenceClass = class(D)
negativeClass = class(N)
```

```
sumClass =
    'pdvar'
sumSize =
```

```
     2     2
sumDegree =
     1
sumContainsDecision =
  logical
   1
differenceClass =
    'pdvar'
negativeClass =
    'pdvar'
```



A nonlinear symbolic operand such as `x*x` is rejected. Rate-vertex rows from `rhodiff` may be added to compatible ordinary expressions; the ordinary rows are broadcast over the active rate vertices.

5.6.2 Multiplication By Known Or Numeric Data

Multiplication is supported only when decision dependence occurs on at most one side. A coefficient-backed `pdmat` or finite real numeric matrix may multiply a `pdvar` from either side. A scalar `pdvar` may scale a known matrix. Two decision-bearing factors, including `P*Q` and `P*x` for a decision `sdpvar x`, are nonlinear and rejected.

Syntax

Call forms

```
R = A * P
R = P * A
R = M * P
R = P * M
```

The first example shows numeric left and right matrix products without exposing incidental YALMIP variable identifiers.

Example

```
yalmip('clear')
P = pdvar(2, 1, {[0 1]}, "full");
L = [1 2] * P;
R = P * [4 5];
S = 3 * P;
leftSize = size(L)
rightSize = size(R)
scaledSize = size(S)
degrees = [L.Degree R.Degree S.Degree]
```

```
leftSize =
     1     1
rightSize =
     2     2
scaledSize =
     2     1
degrees =
     1     1     1
```

A known parameter-dependent factor adds its Bernstein degree through ordered coefficient convolution. On one cell, a quadratic decision and affine known factor produce

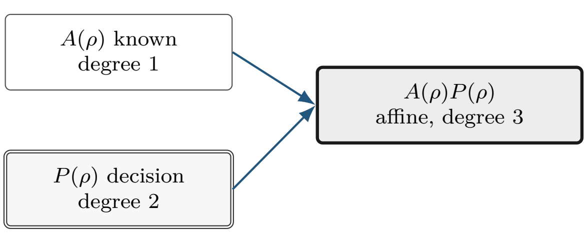
$$(PA)[k] = \sum_{i+j=k} \frac{\binom{2}{i}\binom{1}{j}}{\binom{3}{k}} P[i]A[j], \qquad k = 0, 1, 2, 3.$$

The same backend rule is documented at `bernProd` in section 3.6.4 and derived step by step in Appendix A.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric", Degree=2);
A = pdmat({[0 1]}, {eye(2), 2*eye(2)}, Degree=1);
L = A * P;
R = P * A;
knownLeftClass = class(L)
knownLeftDegree = L.Degree
knownRightClass = class(R)
knownRightDegree = R.Degree
coefficientsPerCell = numel(L.coeffs(1))
```

```
knownLeftClass =
    'pdvar'
knownLeftDegree =
     3
knownRightClass =
    'pdvar'
knownRightDegree =
     3
coefficientsPerCell =
     4
```



A derivative object may be multiplied by numeric data or a matching-grid known object while preserving every rate row. Products with rate dependence on both sides, products of a derivative with an ordinary decision expression, and products involving metadata-only rate dependence are rejected.

Two decision-bearing factors are also rejected: their product would be quadratic in YALMIP decision coefficients. The supported boundary is numeric or known `pdmat` data multiplying one affine `pdvar` expression on either side, with the written matrix order preserved.

Remarks

Incompatible affine operands raise `pdvar:InvalidAddition` or `pdvar:InvalidSubtraction`. Bad dimensions, two decision-bearing factors, nonlinear symbolic data, or unsupported rate combinations raise `pdvar:InvalidMultiplication`. Incompatible parameter bounds raise `pdvar:MixedGrid`; a function-only `pdmat` operand raises `pdvar:FunctionOnlyAlgebra`.

5.7 Matrix Methods

Every method below transforms each local affine YALMIP payload while preserving the physical grid, degree, and compatible rate rows. The examples report stable classes, dimensions, and metadata rather than YALMIP's internal variable numbers.

5.7.1 Shape, Equality, And MATLAB Protocols

Shape queries describe the matrix payload. `isequal` is an exact identity check: grid, degree, matrix size, metadata, and local symbolic payloads must already match; it does not merge or elevate operands.

Syntax

Call forms

```
sz = size(P)
d = size(P, dim)
n = height(P)
n = width(P)
n = length(P)
n = numel(P)
n = ndims(P)
tf = isequal(P, Q, ...)
idx = end(P, k, n)
n = numArgumentsFromSubscript(P, S, ctx)
```

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
matrixSize = size(P)
rowCount = height(P)
columnCount = width(P)
longestDimension = length(P)
elementCount = numel(P)
dimensionCount = ndims(P)
equalsUnaryPlus = isequal(P,+P)
```

```
matrixSize =
     2     3
rowCount =
```

```
     2
columnCount =
     3
longestDimension =
     3
elementCount =
     6
dimensionCount =
     2
equalsUnaryPlus =
  logical
   1
```

`end`, `subsref`, `subsasgn`, and `numArgumentsFromSubscript` support the two-index and dot-access forms shown in section 5.8; they do not expose physical-cell storage.

5.7.2 Transpose And Conjugate Transpose

Both forms transpose every local affine payload without breaking the YALMIP coefficient graph.

Syntax

Call forms

```
T = transpose(P)
H = ctranspose(P)
T = P.'
H = P'
```

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
T = P.';
H = P';
transposeSize = size(T)
conjugateTransposeSize = size(H)
transposeClass = class(T)
conjugateTransposeClass = class(H)
```

```
transposeSize =
     3     2
conjugateTransposeSize =
     3     2
transposeClass =
    'pdvar'
conjugateTransposeClass =
    'pdvar'
```

5.7.3 Vectorization, Reshape, And Squeeze

`vec` uses MATLAB column-major order. `reshape` accepts two positive dimensions with at most one inferred `[]` dimension and preserves the element count. `squeeze` retains the two-dimensional package convention.

Syntax

Call forms

```
V = vec(P)
R = reshape(P, [m n])
R = reshape(P, m, n)
S = squeeze(P)
```

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
V = vec(P);
R = reshape(P, [3 2]);
S = squeeze(P);
vectorSize = size(V)
reshapeSize = size(R)
squeezeSize = size(S)
```

```
vectorSize =
     6     1
reshapeSize =
     3     2
squeezeSize =
     2     3
```

Malformed or size-changing reshape requests raise `pdvar:InvalidReshape`.

5.7.4 Diagonals And Trace

`diag` extracts a diagonal or constructs a diagonal matrix from a vector; the selected offset may not be empty. `trace` returns a scalar decision expression.

Syntax

Call forms

```
d = diag(P)
d = diag(P, k)
D = diag(v)
D = diag(v, k)
t = trace(P)
```

Example

```
yalmip('clear')
P = pdvar(3, {[0 1]}, "full");
d = diag(P);
D = diag(d);
t = trace(P);
diagonalSize = size(d)
rebuiltSize = size(D)
traceSize = size(t)
```

```
diagonalSize =
     3     1
rebuiltSize =
     3     3
traceSize =
     1     1
```

Invalid offsets or empty selections raise `pdvar:InvalidDiag`.

5.7.5 Triangular Parts And Reductions

Triangular methods zero the complementary part of every coefficient. Reductions act on matrix dimensions; `"all"` is available for `sum` and `mean`.

Syntax

Call forms

```
L = tril(P)
L = tril(P, k)
U = triu(P)
U = triu(P, k)
S = sum(P)
S = sum(P, dim)
S = sum(P, "all")
M = mean(P)
M = mean(P, dim)
M = mean(P, "all")
C = cumsum(P)
C = cumsum(P, dim)
```

First form the two triangular views.

Example

```
yalmip('clear')
P = pdvar(3, {[0 1]}, "full");
L = tril(P);
U = triu(P);
lowerSize = size(L)
upperSize = size(U)
degrees = [L.Degree U.Degree]
```

```
lowerSize =
     3     3
upperSize =
     3     3
degrees =
     1     1
```

Then reduce a rectangular expression.

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
rowSum = sum(P, 2);
allMean = mean(P, "all");
running = cumsum(P, 2);
rowSumSize = size(rowSum)
allMeanSize = size(allMean)
runningSize = size(running)
```

```
rowSumSize =
     2     1
allMeanSize =
     1     1
runningSize =
     2     3
```

Invalid triangular offsets raise `pdvar:InvalidTriangularPart`. Invalid reduction calls raise `pdvar:InvalidSum`, `pdvar:InvalidMean`, or `pdvar:InvalidCumsum`.

5.7.6 Reordering And Rotation

These methods rearrange matrix entries inside every symbolic coefficient; they do not reverse the parameter grid.

Syntax

Call forms

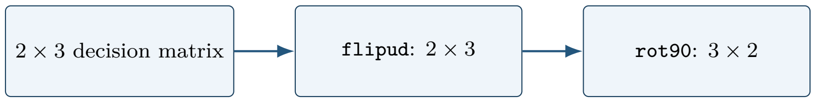
```
Q = flip(P)
Q = flip(P, dim)
Q = flipud(P)
Q = fliplr(P)
Q = rot90(P)
Q = rot90(P, k)
```

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
F = flipud(P);
R = rot90(P);
flippedSize = size(F)
rotatedSize = size(R)
```

```
flippedClass = class(F)
rotatedClass = class(R)
```

```
flippedSize =
     2     3
rotatedSize =
     3     2
flippedClass =
    'pdvar'
rotatedClass =
    'pdvar'
```



Invalid calls raise `pdvar:InvalidFlip` or `pdvar:InvalidRot90`.

5.7.7 Concatenation And Block Assembly

Assembly accepts compatible `pdvar`, coefficient-backed `pdmat`, affine `sdpvar`, and finite real numeric blocks. It aligns grids and degrees while preserving an affine expression. `cat` supports only matrix dimensions 1 and 2.

Syntax

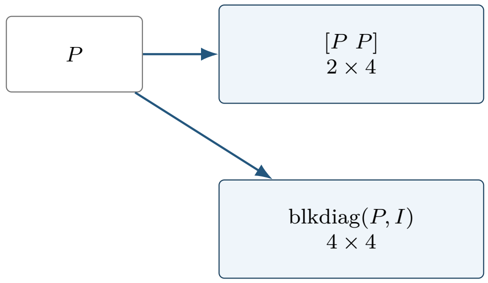
Call forms

```
H = horzcat(P, Q, ...)
V = vertcat(P, Q, ...)
H = [P, Q]
V = [P; Q]
C = cat(dim, P, Q, ...)
D = blkdiag(P, Q, ...)
```

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "full");
H = [P, P];
V = [P; P];
D = blkdiag(P, eye(2));
horizontalSize = size(H)
verticalSize = size(V)
blockDiagonalSize = size(D)
```

```
horizontalSize =
     2     4
verticalSize =
     4     2
blockDiagonalSize =
     4     4
```



Incompatible blocks or rate metadata raise `pdvar:InvalidConcatenation`; unsupported dimensions raise `pdvar:UnsupportedCatDimension`; invalid block-diagonal inputs raise `pdvar:InvalidBlkdiag`. Incompatible bounds raise `pdvar:MixedGrid`; function-only known blocks raise `pdvar:FunctionOnlyAlgebra`.

5.7.8 Repetition

`repmat` tiles the matrix payload while retaining the parameter metadata. It accepts a positive scalar, two positive counts, or a two-element count vector.

Syntax

Call forms

```
Q = repmat(P, m)
Q = repmat(P, m, n)
Q = repmat(P, [m n])
```

Example

```
yalmip('clear')
P = pdvar(1, 2, {[0 1]}, "full");
Q = repmat(P, [2 2]);
repeatedSize = size(Q)
repeatedDegree = Q.Degree
containsDecision = Q.ContainsDecision
```

```
repeatedSize =
     2     4
repeatedDegree =
     1
containsDecision =
  logical
   1
```

Invalid repetition shapes raise `pdvar:InvalidRepmat`.

5.8 Indexing And Assignment

Parenthesis indexing selects a matrix block from every symbolic coefficient. Assignment writes a compatible numeric, `pdvar`, coefficient-backed `pdmat`, or affine `sdpvar` block into that position. It does not select one physical cell.

Syntax

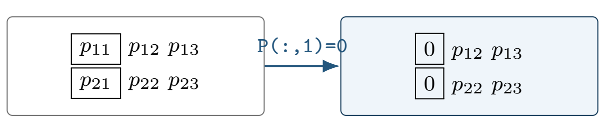
Call forms

```
Q = P(rows, cols)
Q = P(1:end, end)
value = P.PropertyName
P(rows, cols) = numericValue
P(rows, cols) = Q
P(rows, cols) = A
P(rows, cols) = X
```

Example

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
lastCol = P(:, end);
P(:, 1) = 0;
selectedSize = size(lastCol)
assignedSize = size(P)
assignedClass = class(P)
assignedDegree = P.Degree
containsDecision = P.ContainsDecision
```

```
selectedSize =
     2     1
assignedSize =
     2     3
assignedClass =
    'pdvar'
assignedDegree =
     1
containsDecision =
  logical
   1
```



Invalid selectors raise `pdvar:InvalidSubscript`. Linear indexing, deletion, or non-parenthesis assignment raises `pdvar:UnsupportedAssignment`; incompatible or nonlinear right-hand sides raise `pdvar:InvalidAssignment`.

5.9 `value`

Syntax

Call forms

```
A = value(P)
rows = value(rhodiff(P))
```

`value` converts assigned `pdvar` coefficients into known, coefficient-backed `pdmat` data. An ordinary expression returns one `pdmat`, preserving its grid, matrix size, degree, local coefficient order, and continuity metadata. A rate-vertex expression created by `rhodiff` returns a 1-by-2^{ℓ} cell array of `pdmat` objects in `helper.combRows(RateBounds)` lower/upper vertex order. The returned `pdmat` objects do not carry rate metadata. Every symbolic coefficient must have an assigned finite real YALMIP value; otherwise the method raises `pdvar:UnassignedValue`.

Examples and output

Example

```
yalmip('clear')
P = pdvar(1, [0 1], Degree=2);
c = P.coeffs(1);
assign([c{:}], [1 2 4]);
A = value(P);
assignedCoefficients = A.coeffs(1)
```

```
assignedCoefficients =
    {[1]}    {[2]}    {[4]}
```

5.10 Comparisons To `pdlmi`

The `le`, `ge`, and `eq` overloads implement `<=`, `>=`, and `==`. Inequalities select either semidefinite or entry-wise meaning once for the complete original residual; equality is always direct coefficient equality.

Syntax

Call forms

```
C = P <= 0
C = P >= 0
C = lhs <= rhs
C = lhs >= rhs
C = lhs == rhs
```

Arguments

- `rhs` may be compatible numeric, coefficient-backed `pdmat`, affine `sdpvar`, or `pdvar` data. Ordinary operands use the same grid refinement, degree alignment, and promotion as subtraction.
- For `<=` and `>=`, every coefficient in every physical cell and rate row is scanned before assembly. The complete relation is semidefinite only if every original coefficient is square Hermitian. Numeric Hermitian testing uses the inclusive rule $\|A-A^{\mathsf{T}}\|_\infty \leq 10^{-10}$; symbolic coefficients use YALMIP's `ishermitian`. If any coefficient fails, the complete inequality is entry-wise and warning `pdlmi:ElementwiseInequality` is issued once. Modes are never mixed coefficient by coefficient or entry by entry.
- `==` accepts rectangular and nonsymmetric residuals without that warning. It equates corresponding coefficient entries directly. Ordinary rows compare with ordinary rows; derivative rows compare only with compatible derivative rows.
- `C` is a `pdlmi` wrapper. A comparison assembles constraints but does not solve an optimization problem.

Examples and output

Comparison creates a `pdlmi` wrapper with one stored constraint for each physical cell and local Bernstein coefficient.

Example

```
yalmip('clear')
P = pdvar(2, {[0 0.5 1]}, "symmetric");
Cneg = P <= 0;
negativeWrapperClass = class(Cneg)
negativeConstraintCount = numel(Cneg.Constraints)
Cpos = P >= 0;
positiveWrapperClass = class(Cpos)
positiveConstraintCount = numel(Cpos.Constraints)
```

```
negativeWrapperClass =
    'pdlmi'
negativeConstraintCount =
     4
positiveWrapperClass =
    'pdlmi'
positiveConstraintCount =
     4
```

A rectangular residual selects entry-wise inequality globally. Equality is entry-wise by definition and is warning-free.

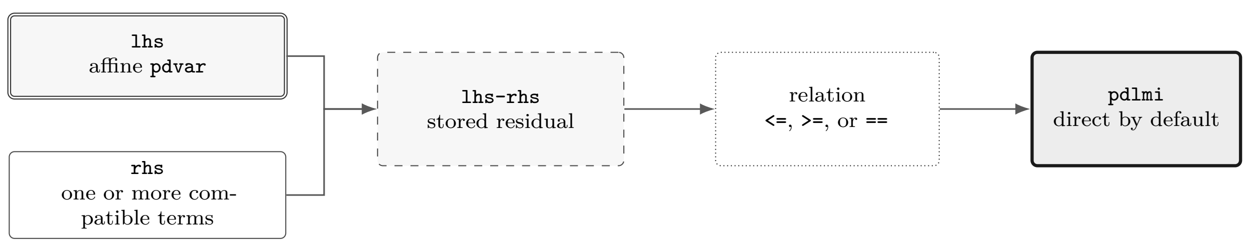
Example

```
yalmip('clear')
V = pdvar(3, 2, {[0 1]}, "full");
warnId = "pdlmi:ElementwiseInequality";
```

```
warnState = warning("off", warnId);
restoreWarning = onCleanup(@() warning(warnState));
Centry = V >= 0;
Ceq = V == sdpvar(3, 2, 'full');
entrywiseConstraintGroups = numel(Centry.Constraints)
equalityConstraintGroups = numel(Ceq.Constraints)
identity = V == V;
identityConstraints = identity.toYalmip()
clear restoreWarning
```

```
entrywiseConstraintGroups =
     2
equalityConstraintGroups =
     2
identityConstraints =
     []
```

Each group above corresponds to one local coefficient and contains six scalar entries in MATLAB column-major order. Under Pólya, Putinar, or Full Box, each entry of an entry-wise inequality receives an independent scalar certificate. Equality is direct-only: constructor certificate options and `applyPolya`, `applyPutinar`, or `applyFullBoxPreorder` raise `pdlmi:UnsupportedEqualityCertificate` before validating an order. Mixed ordinary/derivative row kinds raise `pdvar:InvalidEqualityRows`. Use overloaded `==` to build a modeling equality; use `isequal` only for MATLAB structural comparison.



5.11 Remarks

1. Constructor-created `pdvar` objects support every finite nonnegative integer degree. Higher degree increases the number of Bernstein control matrices per cell.
2. Products may contain decision variables on at most one side; nonlinear decision products such as `P * Q` are outside this slice.
3. Function-only `pdmat` operands cannot enter `pdvar` algebra unless they were constructed with explicit Bernstein evidence.
4. `rhodiff` of an existing rate-vertex derivative expression is not part of the current public workflow.

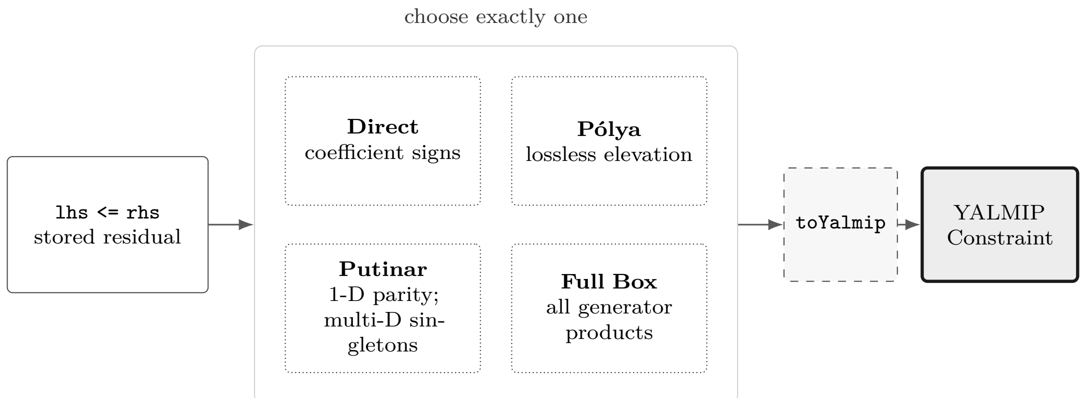
5.11.1 See Also

`pdvar`, `rhodiff`, `bernsteinTable`, `pdlmi`, `pdmat`.

6

pdlmi Reference

`pdlmi` stores YALMIP constraints assembled from `pdvar` expressions. Square Hermitian inequality families use semidefinite constraints; other inequalities use entry-wise scalar constraints; equality uses direct coefficient equations. Direct assembly is the default, while cell-local Pólya elevation, a dimension-aware Putinar selector, and the full box Bernstein–Gram preordering are explicit inequality alternatives. It is the package boundary between Bernstein objects and ordinary YALMIP solver calls.



The four certificate paths are alternatives. Selecting Pólya, Putinar, or full-box rebuilding replaces, rather than compounds, the previous selection.

6.1 Lookup

Task	Entry
Construct wrapper	`pdlmi`, `<=`/`>=`/`==`
Count constraints	direct coefficient constraints
Select Pólya elevation	`UsePolya`, `PolyaDegree`, and `applyPolya`
Select Putinar certificate	`UsePutinar`, `PutinarOrder`, and `applyPutinar`
Select full box preordering	`UseFullBoxPreorder`, `FullBoxOrder`, and `applyFullBoxPreorder`
Export to YALMIP	`toYalmip`
Solve	solver-facing use, Examples

6.2 Properties

All `pdlmi` properties have private set access and are readable with dot notation. `Constraints` contains the assembled YALMIP entries; `Residual` and `Relation` retain the original comparison for rebuilding. The remaining properties record the selected certificate state.

Property	Meaning
`Constraints`	Stored finite YALMIP constraint entries.
`Residual`, `Relation`	Original `pdvar` residual and one of `"<="`, `">="`, or `"=="`.
`UsePolya`, `PolyaDegree`	Pólya selector and nonnegative degree increment.
`UsePutinar`, `PutinarOrder`	Putinar selector and absolute Gram order.
`UseFullBoxPreorder`, `FullBoxOrder`	Full-box selector and absolute Gram order.

For example, `C.Relation`, `C.Residual.Degree`, and `C.UsePutinar` inspect a wrapper. Apply methods return a rebuilt value; assigning these properties is unsupported.

6.3 Construction And Comparisons

Syntax

Call forms

```
C = pdlmi(expr, relation)
C = pdlmi(expr, "==")
C = pdlmi(expr, relation, "UsePolya")
C = pdlmi(expr, relation, UsePolya=true, PolyaDegree=d)
C = pdlmi(expr, relation, "UsePolya", "PolyaDegree", d)
C = pdlmi(expr, relation, "UsePutinar")
C = pdlmi(expr, relation, UsePutinar=true, PutinarOrder=r)
C = pdlmi(expr, relation, PutinarOrder=r)
C = pdlmi(expr, relation, "UseFullBoxPreorder")
C = pdlmi(expr, relation, UseFullBoxPreorder=true)
C = pdlmi(expr, relation, FullBoxOrder=r)
C = pdlmi(expr, relation, UseFullBoxPreorder=true, FullBoxOrder=r)
C = pdlmi(expr, relation, UseFullBoxPreorder=false, FullBoxOrder=r)
C = lhs <= rhs
C = lhs >= rhs
C = lhs == rhs
```

Arguments

- `expr` is a `pdvar` residual; `relation` is exactly `"<="`, `">="`, or `"=="`. Inequalities are classified globally from the original coefficient family. Equality is entry-wise direct coefficient assembly and accepts rectangular or nonsymmetric residuals.
- `UsePolya` is a logical scalar option with default `false`. Its bare form, `"UsePolya"`, and `UsePolya=true` without a degree select increment one.
- `PolyaDegree=d` is a finite nonnegative integer scalar. It elevates the residual by `d` in every parameter direction before assembling constraints. Supplying it without `UsePolya` enables

Pólya and issues `pdlmi:ImplicitUsePolya`; `UsePolya=false` with positive `d` raises `pdlmi:ConflictingPolyaOptions`.

- `UsePutinar` is a logical scalar option with default `false`. Its bare form and the explicit value `true` select the dimension-appropriate Putinar certificate at the smallest admissible order.
- `PutinarOrder=r` is a finite nonnegative integer scalar and is an absolute Bernstein Gram order, not a degree increment. For residual degree m, its default and minimum are $\lfloor m/2 \rfloor$ in one parameter and $\lceil m/2 \rceil$ in two or more parameters. Supplying the order without `UsePutinar` enables Putinar assembly. Pairing `UsePutinar=false` with any explicit order, including zero, raises `pdlmi:ConflictingPutinarOptions`.
- `UseFullBoxPreorder` is a logical scalar option with default `false`. Its bare form and the explicit value `true` select full-box assembly at the smallest admissible order.
- `FullBoxOrder=r` is a finite nonnegative integer scalar and is an absolute order, not a degree increment. Supplying it without `UseFullBoxPreorder` enables full-box assembly. When full-box assembly is enabled, the minimum is $\lfloor m/2 \rfloor$ for a one-parameter residual of degree m, and $\lceil m/2 \rceil$ for two or more parameters. Explicitly pairing `UseFullBoxPreorder=false` with `FullBoxOrder=r` suppresses that automatic selection: `C` remains direct and records `r` only as inspectable metadata; the order is not applied to assembly.
- Pólya, Putinar, and full-box selection are mutually exclusive and apply only to inequalities. Any recognized certificate option on an equality raises `pdlmi:UnsupportedEqualityCertificate`. Enabling more than one inequality family raises `pdlmi:ConflictingRelaxations`.
- `C` is direct coefficient-wise assembly unless one opt-in inequality mode is selected. Comparisons such as `lhs >= rhs` are the ordinary user-facing entry point and accept compatible numeric, `pdmat`, affine `sdpvar`, and `pdvar` right-hand sides.

The inspectable properties `Constraints`, `Residual`, and `Relation` expose the assembled finite constraints and the original comparison data used for rebuilding. `UsePolya`, `PolyaDegree`, `UsePutinar`, `PutinarOrder`, `UseFullBoxPreorder`, and `FullBoxOrder` report the selected mode and order metadata. Their set access is private. Comparison-created direct objects have all three selectors false and all three numeric properties zero. The full-box explicit-false combination described above is the exception: it retains `FullBoxOrder=r` while `UseFullBoxPreorder` remains false and the stored constraints remain direct. Putinar deliberately has no corresponding exception.

Examples and output

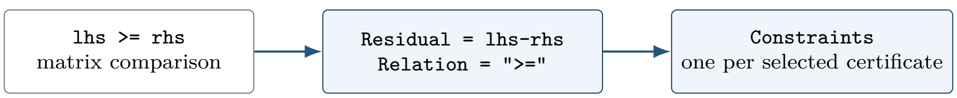
The direct constructor is equivalent to the comparison path when the relation and options are explicit.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric");
C = pdlmi(P, ">=", UsePolya=false, PolyaDegree=0);
wrapperClass = class(C)
constraintCount = numel(C.Constraints)
```

```
usePolya = C.UsePolya
polyaDegree = C.PolyaDegree
usePutinar = C.UsePutinar
putinarOrder = C.PutinarOrder
useFullBox = C.UseFullBoxPreorder
fullBoxOrder = C.FullBoxOrder
```

```
wrapperClass =
    ’pdlmi’
constraintCount =
     2
usePolya =
  logical
   0
polyaDegree =
     0
usePutinar =
  logical
   0
putinarOrder =
     0
useFullBox =
  logical
   0
fullBoxOrder =
     0
```



Remarks

---

1. A non-`pdvar` residual raises `pdlmi:InvalidExpression`; an unsupported relation raises `pdlmi:InvalidRelation`.

2. Malformed, unknown, or duplicate options raise `pdlmi:InvalidOptions`, `pdlmi:UnknownOption`, or `pdlmi:DuplicateOption`; nonlogical selectors use their corresponding `InvalidUse...` identifier.

3. Malformed Pólya increments, Putinar orders, and full-box orders raise, respectively, `pdlmi:InvalidPolyaDegree`, `pdlmi:InvalidPutinarOrder`, and `pdlmi:InvalidFullBoxOrder`. An integer below an active Gram-family minimum raises `pdlmi:PutinarOrderTooLow` or `pdlmi:FullBoxOrderTooLow`.

4. Explicit false plus a Putinar order raises `pdlmi:ConflictingPutinarOptions`; explicit false plus a positive Pólya increment raises `pdlmi:ConflictingPolyaOptions`. The full-box explicit-false form remains direct and retains the supplied order as metadata.

5. An inequality with any nonsquare or non-Hermitian original coefficient is assembled entry-wise and emits `pdlmi:ElementwiseInequality`; it is not rejected. Numeric Hermitian testing accepts equality at the inclusive 10^{-10} infinity-norm tolerance.

6. Equality is direct-only. Every `apply...` method raises `pdlmi:UnsupportedEqualityCertificate` before validating its supplied increment or order.

6.4 Direct Coefficient Assembly

For each physical cell, local Bernstein coefficient, and rate-vertex row if present, `pdlmi` creates one stored YALMIP constraint entry. For an ordinary expression with N_c physical cells and N_b Bernstein coefficients per cell, the number of entries is $N_c N_b$; with N_v rate vertices it is $N_c N_b N_v$. A semidefinite-mode entry is one matrix constraint. An entry-wise-mode entry is the column-major vector of scalar matrix entries compared independently. An equality entry is the corresponding direct coefficient equation and may be rectangular.

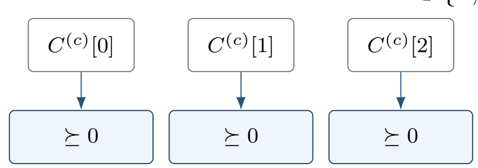
The idea is the Bernstein convex-hull property. After orienting the relation as a positive-semidefinite target

$$S_{\mathbf{c},v}(\boldsymbol{\alpha}) = \sum_{\mathbf{i}} B_{\mathbf{i}}^{m}(\boldsymbol{\alpha}) C^{(\mathbf{c},v)}[\mathbf{i}],$$

the semidefinite direct certificate requires $C^{(\mathbf{c},v)}[\mathbf{i}] \succeq 0$ for every physical cell $\mathbf{c}$, active rate vertex v, and local label $\mathbf{i}$. This is sufficient, not necessary.

For an entry-wise inequality, replace each matrix coefficient by `C(:)` and impose the oriented scalar sign on every component. The global classification is made before Pólya elevation or Gram assembly, so every coefficient and every certificate in one wrapper uses the same mode. Putinar and Full Box create one independent scalar Gram certificate for each column-major entry. Direct equality simply imposes `C(:) == 0`; it does not use a positivity certificate or an implicit tolerance.

The diagram specializes to one parameter, degree two, and one ordinary (non-rate) row; therefore c is a scalar one-based cell index and $i \in \{0, 1, 2\}$.



repeat independently for each cell and rate row

6.5 Opt-In Pólya Assembly And `applyPolya`

With `UsePolya=true`, `pdlmi` first elevates the stored residual by `PolyaDegree` in every parameter direction, then constrains every elevated local coefficient and every active rate row. Thus, if the original degree is m, the count is $N_c(m+d+1)^{\ell}$ for ordinary expressions and $N_c(m+d+1)^{\ell} N_v$ for rate-vertex expressions. An increment of zero is valid and has the same coefficient count as direct assembly. For an entry-wise inequality, elevation preserves the global mode and the oriented sign is applied independently to every column-major scalar entry of each elevated coefficient. Equality wrappers reject this method.

Syntax

Call forms

```
Cpolya = C.applyPolya()
Cpolya = C.applyPolya(d)
```

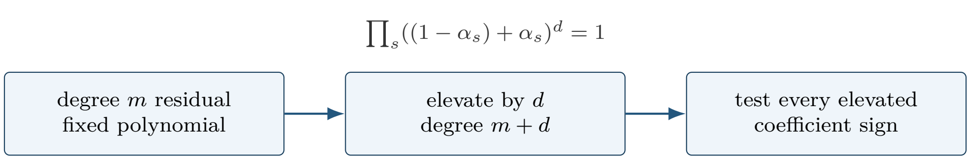
`applyPolya` is a value-class method: it returns a new `pdlmi`, leaves `C` unchanged, and rebuilds from `C.Residual`. Selecting a new increment therefore replaces rather than compounds a previous selection. The no-argument form uses one; an invalid increment raises `pdlmi:InvalidPolyaDegree`.

Examples and output

Example

```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=1);
direct = P >= 0;
polya = direct.applyPolya(2);
directDegree = direct.PolyaDegree
usePolya = polya.UsePolya
polyaDegree = polya.PolyaDegree
constraintCount = numel(polya.Constraints)
```

```
directDegree =
     0
usePolya =
  logical
   1
polyaDegree =
     2
constraintCount =
     4
```



Pólya degree is a uniform representation increment, not a new decision degree, a grid refinement, or a Gram order.

The mathematical background and the limits of fixed-increment coefficient positivity are developed in section B.

6.6 Putinar Selection And `applyPutinar`

For an entry-wise inequality, the constructions below are applied independently to each scalar matrix entry in MATLAB column-major order. Gram variables are not shared between entries. Equality wrappers reject Putinar selection.

The Putinar selector is dimension-aware. With one parameter it deliberately uses the same exact even/odd Markov–Lukács assembly as `applyFullBoxPreorder`. For selected absolute order r, the even form matches at degree $2r$,

$$F = \mathcal{Z}_r^\top Q_0 \mathcal{Z}_r + \alpha(1-\alpha)\mathcal{Z}_{r-1}^\top Q_1 \mathcal{Z}_{r-1},$$

where the second block is omitted at $r = 0$. The odd form matches at degree $2r+1$,

$$F = (1-\alpha)\mathcal{Z}_r^\top Q_L \mathcal{Z}_r + \alpha \mathcal{Z}_r^\top Q_U \mathcal{Z}_r.$$

All present Gram matrices are positive semidefinite. The one-parameter default and minimum are $r = \lfloor m/2 \rfloor$ for residual degree m.

For two or more parameters, the selector instead uses the fixed-order, box-specific singleton-generator Bernstein–Gram quadratic module. After normalizing the relation so that $F(\boldsymbol{\alpha})$ is positive semidefinite, it represents

$$F(\boldsymbol{\alpha}) = S_0(\boldsymbol{\alpha}) + \sum_{s=1}^{\ell} \alpha_s(1-\alpha_s)S_s(\boldsymbol{\alpha}).$$

Only singleton box generators appear in this multivariate branch: products such as $\alpha_1(1-\alpha_1)\alpha_2(1-\alpha_2)$ belong to the full-box preordering, not this module.

Syntax

Call forms

```
Cputinar = C.applyPutinar()
Cputinar = C.applyPutinar(r)
```

Here **r** is an absolute Bernstein Gram order. The no-argument form uses the dimension-dependent minima above; an explicit order must satisfy the same applicable bound. In the multivariate branch, the unweighted block S_0 uses tensor basis degree $r\mathbf{1}$, while S_s uses degree $r\mathbf{1} - \boldsymbol{e}_s$. A nominal block with a negative component is omitted, which occurs for every multiplier at $r = 0$. The target is degree-elevated without changing its represented polynomial, and the multivariate Gram sum is matched exactly at tensor degree $2r$.

Examples and output

Example

```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
putinar = direct.applyPutinar();
directUsesPutinar = direct.UsePutinar
directOrder = direct.PutinarOrder
directConstraintCount = numel(direct.Constraints)
selectedUsesPutinar = putinar.UsePutinar
selectedOrder = putinar.PutinarOrder
selectedConstraintCount = numel(putinar.Constraints)
```

```
directUsesPutinar =
  logical
   0
directOrder =
     0
directConstraintCount =
     3
selectedUsesPutinar =
  logical
   1
selectedOrder =
     1
selectedConstraintCount =
     5
```

For this one-parameter scalar degree-two cell, order one uses the exact even Markov–Lukács form: an unweighted Gram block of dimension two and one generator-weighted block of dimension one, followed by three exact coefficient identities. The result is identical to `direct.applyFullBoxPreorder()`. Each physical cell and each active rate-vertex row receives independent copies of these PSD blocks and identities.

The diagram below specializes to the even one-parameter example above.

The method changes the finite certificate, not the stored residual, decision degree, or physical grid.

`applyPutinar` is a value-class method: it returns a new `pdlmi`, leaves `C` unchanged, and rebuilds from the original `Residual` and `Relation`. Reapplication replaces any earlier Pólya, Putinar, or full-box selection. It adds no implicit margin and makes no solver call.

Remarks

---

1. Malformed orders raise `pdlmi:InvalidPutinarOrder`; an integer below $\lfloor m/2 \rfloor$ in one parameter, or below $\lceil m/2 \rceil$ in two or more parameters, raises `pdlmi:PutinarOrderTooLow`.

2. A malformed selector raises `pdlmi:InvalidUsePutinar`; explicit false plus any supplied order raises `pdlmi:ConflictingPutinarOptions`.

3. Simultaneous relaxation families raise `pdlmi:ConflictingRelaxations`; expression, relation, and option validation in section 6.3 still applies. Global inequality classification is unchanged: the family is semidefinite only when every original coefficient across all cells and rate rows is square Hermitian. Otherwise the complete family uses independent entry-wise scalar Putinar certificates in MATLAB column-major order.

6.7 Full Box Bernstein–Gram Preordering And `applyFullBoxPreorder`

The full-box mode is a cell-local positive-semidefinite representation in the forward local coordinates $\alpha_s \in [0, 1]$. It is a box-specific Bernstein–Gram preordering, not a general Putinar certificate, general-domain SOS interface, or full-space SOS representation. It adds no implicit positivity margin. For an entry-wise inequality, every column-major scalar entry receives an independent copy of the selected Gram construction. Equality wrappers reject full-box selection.

Syntax

Call forms

```
Cbox = C.applyFullBoxPreorder()
Cbox = C.applyFullBoxPreorder(r)
```

The no-argument form selects the smallest admissible absolute order. The explicit form selects `r`; it does not add `r` to the residual degree. Both forms are value-class selectors: they return a new `pdlmi`, leave `C` unchanged, and rebuild from `C.Residual`. Reapplication therefore replaces an earlier full-box order or a Pólya or Putinar selection rather than compounding it.

The idea is to represent the oriented positive target as a sum of PSD Gram forms multiplied by functions that are nonnegative on the local box:

$$S(\boldsymbol{\alpha}) = \sum_{J\subseteq\{1,\ldots,\ell\}} \left(\prod_{s\in J} \alpha_s(1-\alpha_s)\right) \mathcal{Z}_J(\boldsymbol{\alpha})^{\mathsf{T}} Q_J \mathcal{Z}_J(\boldsymbol{\alpha}), \qquad Q_J \succeq 0.$$

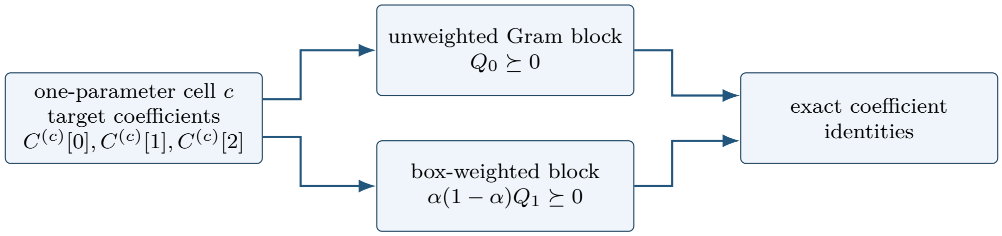
In one parameter the implementation uses the even/odd Markov–Lukács forms, with minimum absolute order $\lfloor m/2 \rfloor$ for residual degree m. For two or more parameters it uses every subset product of the ℓ box generators and minimum order $\lceil m/2 \rceil$. The selected order fixes the dense Bernstein Gram bases; it is not added to m.

Example

```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
box = direct.applyFullBoxPreorder();
directConstraintCount = numel(direct.Constraints)
useFullBox = box.UseFullBoxPreorder
fullBoxOrder = box.FullBoxOrder
fullBoxConstraintCount = numel(box.Constraints)
```

```
directConstraintCount =
     3
useFullBox =
  logical
   1
fullBoxOrder =
     1
fullBoxConstraintCount =
     5
```

At this order the scalar degree-two cell has two PSD Gram constraints followed by three exact Bernstein coefficient identities.



Appendix B derives the one-dimensional parity forms, explains the multivariate subset masks, and distinguishes this implemented box preordering from the multivariate singleton-generator Putinar branch, full-space SOS, and general polynomial parsers. Assembly is independent for every physical cell and every active rate-vertex row; Gram variables are not shared across either boundary. For `>=`, the method represents the residual. For `<=`, it negates the target before forming the positive-semidefinite representation. Within each cell and row, the stored constraints are ordered as all PSD Gram blocks followed by exact equalities matching every Bernstein coefficient. Consequently, the number of stored constraints is not the direct coefficient count.

Remarks

---

1. Malformed orders raise `pdlmi:InvalidFullBoxOrder`; an admissible integer below the minimum raises `pdlmi:FullBoxOrderTooLow`. The constructor rejects malformed selectors with `pdlmi:InvalidUseFullBoxPreorder`.
2. Simultaneous Pólya, Putinar, or full-box selection raises `pdlmi:ConflictingRelaxations`; expression, relation, and option validation in section 6.3 still applies.
3. Global inequality classification is unchanged: the family is semidefinite only when every original coefficient across all cells and rate rows is square Hermitian. Otherwise the complete family uses independent entry-wise scalar Full Box certificates in MATLAB column-major order.

6.8 `toYalmip`

Syntax

Call forms

```
F = toYalmip(C)
F = C.toYalmip()
```

Arguments

`toYalmip` concatenates the stored constraint cells into a YALMIP constraint array for direct, Pólya, Putinar, and full-box objects. It has no package-specific options and does not add a solver wrapper, objective, margin, or diagnostic layer. The output is suitable for `optimize`, concatenation with other YALMIP constraints, or ordinary YALMIP inspection.

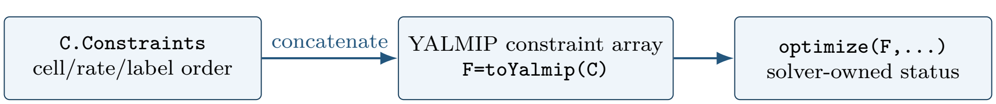
Examples and output

The function-call and dot-call forms return equivalent YALMIP constraint arrays.

Example

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric");
C = P >= 0;
wrapperClass = class(C)
storedConstraintCount = numel(C.Constraints)
F1 = toYalmip(C);
F2 = C.toYalmip();
firstIsConstraint = isa(F1, "lmi") || isa(F1, "constraint")
firstConstraintCount = length(F1)
secondIsConstraint = isa(F2, "lmi") || isa(F2, "constraint")
secondConstraintCount = length(F2)
```

```
wrapperClass =
    'pdlmi'
storedConstraintCount =
     2
firstIsConstraint =
  logical
   1
firstConstraintCount =
     2
secondIsConstraint =
  logical
   1
secondConstraintCount =
     2
```



After this boundary, the package is no longer assembling Bernstein data; YALMIP owns solver choice, diagnostics, and optimization status.

6.9 See Also

`pdvar`, `rhodiff`, `applyPolya`, `applyPutinar`, `applyFullBoxPreorder`, `toYalmip`, `optimize`.

6.10 Solver-Facing Use

pdlmi does not own solver settings or wrap optimize. After toYalmip, use ordinary YALMIP:

Call forms

```
opts = sdpsettings("solver", solverName, "verbose", 0);
sol = optimize([C1.toYalmip, C2.toYalmip], objective, opts);
```

7

Cross-Class Helper Reference

The `+helper` namespace contains exactly nine implementation utilities whose ownership genuinely crosses class boundaries. They are developer-facing building blocks, not an alternative modeling API: ordinary users should enter through `pdbase`, `pdmat`, `pdvar`, and `pdlmi`. Their signatures and current consumers are documented below so maintainers can distinguish shared contracts from class-local implementation details.

7.1 `helper.bernTbl`

Purpose. Build the shared detailed or one-line Bernstein inspection table used by `pdmat` and `pdvar`.

Syntax

Call forms

```
T = helper.bernTbl(obj, errId, valFcn, exprFcn, rateVerts)
T = helper.bernTbl(..., cellSubs)
T = helper.bernTbl(..., "oneLine")
```

Description

`obj` supplies `cells`, `lbls`, and `coeffs`; `valFcn` maps each coefficient to a table value and `exprFcn` maps it to expression text. `rateVerts` is empty for ordinary data or has one row per active rate vertex. The optional `cellSubs` selects one physical cell; `"oneLine"` returns one expression per cell and rate row.

Examples and output

Example

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
T = helper.bernTbl(A, "demo:Invalid", @(x) x, ...
    @(x) string(mat2str(x)), [], "oneLine")
```

```
T =
    CellSubscript          Expression
    _____________    ______________________
        {[1]}        "(1-alpha)*1 + alpha*2"
```

Remarks

Only one cell selector and the case-insensitive `"oneLine"` option are accepted. Invalid selectors, unknown text, repeated selectors, or rate-row count mismatches raise the caller-owned `errId`, preserving the public class error namespace.

7.2 `helper.cellGet`

Purpose. Read one flat coefficient leaf from a nested physical-cell tree.

Syntax

Call forms

```
leaf = helper.cellGet(vals, subs)
```

Description

`vals` is addressed as `vals{i1}{i2}...{i_ell}` and `subs` supplies one index per level. The returned `leaf` is the selected flat coefficient cell or rate-row table; no copy, reshaping, or multi-index cell interpretation is introduced.

Examples and output

Example

```
vals = {{{10, 20}}, {{30, 40}}};
leaf = helper.cellGet(vals, [2 1])
```

```
leaf =
  1x2 cell array
    {[30]}    {[40]}
```

Remarks

The helper deliberately relies on ordinary MATLAB cell indexing. Callers validate the subscript length and bounds before traversal.

7.3 `helper.chk`

Purpose. Apply shared validation predicates while retaining a caller-owned error identifier and message.

Syntax

Call forms

```
val = helper.chk(val, errId, msg, tags...)
val = helper.chk(val, errId, msg, tags..., Name, Value)
```

Description

Predicate tags are `numeric`, `real`, `cell`, `struct`, `nonempty`, `scalar`, `vector`, `matrix`, `finite`, `integer`, `positive`, `nonnegative`, `increasing`, and `rowbounds`. Named tests are `Size`, `Numel`, `MinNumel`, `Min`, and `Max`. Successful validation returns `val` unchanged.

Examples and output

Example

```
validated = helper.chk([1 2], "demo:BadValue", "bad value", ...
    "numeric", "real", "finite", "Size", [1 2])
```

```
validated =
     1     2
```

Remarks

A failed data predicate raises `errId`. An unknown predicate, or a named test without a value, raises `helper:InvalidValidatorCall`; these errors signal a validator-programming defect rather than bad caller data.

7.4 `helper.combRows`

Purpose. Enumerate Cartesian combinations in the ordering shared by physical cells, local labels, and rate vertices.

Syntax

Call forms

```
rows = helper.combRows(vecs)
```

Description

`vecs` is a cell array containing one vector per tensor axis. `rows` contains every Cartesian combination, with earlier dimensions varying more slowly.

Examples and output

Example

```
rows = helper.combRows({0:1, 10:11})
```

```
rows =
     0    10
     0    11
     1    10
     1    11
```

Remarks

This ordering is part of the storage contract. Reordering the rows would misalign tensor labels, physical-cell traversal, and derivative rate vertices.

7.5 `helper.isZero`

Purpose. Prove the distinct zero notions used by known-data and affine algebra fast paths.

Syntax

Call forms

```
tf = helper.isZero(val, "num")
tf = helper.isZero(val, "add", matrixSize)
tf = helper.isZero(vals, "vals")
tf = helper.isZero(obj, "obj")
```

Description

`"num"` requires a nonempty finite real numeric zero; `"add"` additionally enforces scalar expansion or the supplied matrix shape; `"vals"` recursively checks nested, symbolic, and rate-structured payloads; and `"obj"` accepts coefficient-backed `pdmat` or `pdvar` evidence.

Examples and output

Example

```
numericZero = helper.isZero(zeros(2), "add", [2 2])
A = pdmat({[0 1]}, {0, 0}, Degree=1);
objectZero = helper.isZero(A, "obj")
```

```
numericZero =
  logical
   1
objectZero =
  logical
   1
```

Remarks

Function-only `pdmat` placeholders are not coefficient evidence and therefore do not prove object zero. Invalid modes raise `helper:InvalidZeroMode`; the wrong mode-specific arity raises `helper:InvalidZeroCall`.

7.6 `helper.mapVals`

Purpose. Map one payload function across every leaf of a nested `LocalValues` tree.

Syntax

Call forms

```
mapped = helper.mapVals(vals, fcn, grid)
```

Description

`grid` determines the nested physical-cell shape and `fcn` is applied with `cellfun` to every payload in every selected leaf. The result retains the original physical-cell tree and coefficient ordering.

Examples and output

Example

```
vals = {{{1, 2}}, {{3, 4}}};
mapped = helper.mapVals(vals, @(x) 10*x, {[0 1 2], [0 1]});
secondLeaf = helper.cellGet(mapped, [2 1])
```

```
secondLeaf =
  1x2 cell array
    {[30]}    {[40]}
```

Remarks

The mapping preserves nesting but does not validate payload semantics; the owning class remains responsible for matrix sizes, symbolic affinity, and rate-row structure.

7.7 helper.matSubs

Purpose. Normalize supported two-dimensional payload subscripts into explicit row and column index vectors.

Syntax

Call forms

```
[rows, cols] = helper.matSubs(subs, sz, errId)
```

Description

`subs` is a two-element cell array. Each element may be a finite positive integer vector, a matching logical vector, or `":"`; `sz=[m n]` supplies the available matrix size. Errors use the caller-owned `errId`.

Examples and output

Example

```
[rows, cols] = helper.matSubs({":", 2}, [3 4], "demo:BadSub")
```

```
rows =
     1     2     3
cols =
     2
```

Remarks

The helper rejects linear indexing, empty or out-of-range numeric indices, and logical selectors whose lengths do not match the indexed dimension. It never permits array growth.

7.8 `helper.mkGrid`

Purpose. Validate tensor-grid vectors and construct the shared `GridInfo` record.

Syntax

Call forms

```
info = helper.mkGrid(grid)
info = helper.mkGrid(grid, owner)
```

Description

`grid` is a nonempty cell array of finite, real, strictly increasing vectors with at least two nodes each. `owner` defaults to `"pdbase"` and prefixes validation identifiers. The result contains `Vectors`, Cartesian-product `Points`, per-axis `Bounds`, and `NumNodes`.

Examples and output

Example

```
info = helper.mkGrid({[0 1], [10 20]}, "demo");
bounds = info.Bounds
points = info.Points
```

```
bounds =
     0     1
    10    20
points =
     0    10
     0    20
     1    10
     1    20
```

Remarks

Malformed containers use `owner + ":InvalidGrid"`; malformed individual axes use `owner + ":InvalidGridVector"`. Cell counts are derived by consumers and are not duplicated in `GridInfo`.

7.9 `helper.mkNest`

Purpose. Allocate recursive physical-cell storage and construct each leaf from its complete cell subscript.

Syntax

Call forms

```
vals = helper.mkNest(nCell, mkLeaf)
vals = helper.mkNest(nCell, mkLeaf, prefix)
```

Description

`nCell` gives the positive cell count per parameter direction and `mkLeaf` receives one completed one-based subscript row. `prefix` is recursion state used internally; ordinary callers omit it. The result is addressed as `vals{i1}{i2}...{i_ell}`.

Examples and output

Example

```
vals = helper.mkNest([2 1], @(subs) {sum(subs)});
secondLeaf = helper.cellGet(vals, [2 1])
```

```
secondLeaf =
  1x1 cell array
    {[3]}
```

Remarks

`mkLeaf` owns the leaf payload. The helper supplies deterministic physical-cell subscripts but does not infer coefficient count, matrix size, or rate semantics.

7.10 Class-Private Helpers

Single-class mechanics remain beside their owner rather than in `+helper`. The private area of `pdbase` owns coefficient-tree validation. The private area of `pdmat` owns source fitting, continuity, grid conversion, matrix scanning, and algebraic wrapping. The private area of `pdvar` owns affine and rate-row packing, product routing, value-tree zipping, and symbolic wrapping. The private area of `pdlmi` owns direct coefficient, Putinar, and full-box assembly, Gram conversion, and order validation. These functions have no supported direct call form. Their placement is an ownership boundary: a helper should move into a class when only that class can meaningfully use it, and should enter `+helper` only after a real cross-class contract exists.

8

Examples

8.1 Deterministic Putinar And Full-Box Selection

This example selects default Putinar and full-box certificates and an explicit full-box order without solving an optimization problem. The reported values depend only on the public assembly contract, not on incidental YALMIP decision-variable identifiers.

Example

```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
putinarDefault = direct.applyPutinar();
boxDefault = direct.applyFullBoxPreorder();
boxHigher = boxDefault.applyFullBoxPreorder(2);
polya = direct.applyPolya(1);
boxFromPolya = polya.applyFullBoxPreorder();
putinarFromBox = boxHigher.applyPutinar();
Fbox = toYalmip(boxDefault);

directState = [direct.UseFullBoxPreorder, direct.FullBoxOrder, ...
    numel(direct.Constraints)]
defaultState = [boxDefault.UseFullBoxPreorder, boxDefault.FullBoxOrder, ...
    numel(boxDefault.Constraints)]
putinarState = [putinarDefault.UsePutinar, putinarDefault.PutinarOrder, ...
    numel(putinarDefault.Constraints)]
higherState = [boxHigher.FullBoxOrder, numel(boxHigher.Constraints)]
replacementState = [boxFromPolya.UsePolya, ...
    boxFromPolya.UseFullBoxPreorder, boxFromPolya.FullBoxOrder]
putinarReplacementState = [putinarFromBox.UsePutinar, ...
    putinarFromBox.UseFullBoxPreorder, putinarFromBox.PutinarOrder]
exportedConstraintCount = length(Fbox)
```

```
directState =
     0     0     3
defaultState =
     1     1     5
putinarState =
     1     1     5
higherState =
     2     7
replacementState =
     0     1     1
putinarReplacementState =
     1     0     1
exportedConstraintCount =
     5
```

For the degree-two scalar residual, order one uses two PSD Gram blocks followed by three exact coefficient identities. Order two matches a degree-four target and stores two PSD blocks followed by five identities. The original `direct` object remains unchanged, and the selection made from `polya` replaces Pólya rather than elevating its already assembled constraints. Likewise, selecting Putinar from `boxHigher` rebuilds from the original residual and clears the full-box selection.

8.2 Parameter-Dependent Lyapunov Solver Smoke

The first solver smoke test in `+tests/+pdlmi/test_solver_smoke.m` contrasts a constant Lyapunov matrix with a degree-one, parameter-dependent one. It solves the same decay and positivity constraints in both cases. The degree-one solution is then materialized as known `pdmat` data and plotted. `pdvar` is a decision-expression class and does not itself provide `plot`; applying `value` to its local YALMIP payloads after a successful solve supplies the numeric Bernstein coefficients for `pdmat`.

Example

```
yalmip('clear')
grid = {[0 1]};
A = pdmat(grid, @(x) (1 - x) * [-1 -1; 1 -1] + ...
    x * [-1 -10; 0.1 -1], Degree=1);
solver = 'lmilab';
if exist('mosekopt', 'file') ~= 0
    solver = 'mosek';
end
opts = sdpsettings('solver', solver, 'verbose', 0);

Pc = pdvar(2, grid, Degree=0);
CconstantDecay = Pc * A + A' * Pc <= -1e-10 * eye(2);
CconstantPositive = Pc >= eye(2);
solConstant = optimize([CconstantDecay.toYalmip, ...
    CconstantPositive.toYalmip], [], opts);

Pd = pdvar(2, grid);
CdependentDecay = Pd * A + A' * Pd <= -1e-10 * eye(2);
CdependentPositive = Pd >= eye(2);
solDependent = optimize([CdependentDecay.toYalmip, ...
    CdependentPositive.toYalmip], [], opts);
assert(solDependent.problem == 0, solDependent.info)

Pplot = pdmat(grid, ...
    {cellfun(@value, Pd.LocalValues{1}, UniformOutput=false)}, ...
    Degree=Pd.Degree);
figure(Name="Parameter-dependent Lyapunov matrix")
plot(Pplot, SamplesPerCell=40, LineWidth=1.5)
title("Solved parameter-dependent Lyapunov matrix")
```

The figure contains one curve for each entry of the solved 2-by-2 matrix $P(\rho)$, sampled on the parameter interval. It is a diagnostic visualization of the feasible solution, not a certificate beyond the coefficient-wise constraints supplied to the solver. The constant-case result is retained to exercise the comparison in the smoke test; only a MOSEK solve is used there to certify its intended infeasibility distinction.

8.3 Rate-Dependent Block LMI Solver Smoke

This example is adapted from `+tests/+pdlmi/test_solver_smoke.m`. The goal is to find a parameter-dependent $P(\rho)$ and scalar γ such that, on every Bernstein coefficient and rate vertex,

$$\begin{bmatrix} \dot{P} + PA + A^\top P & PB & C^\top \\ B^\top P & -\gamma I & D^\top \\ C & D & -\gamma I \end{bmatrix} \preceq 0, \qquad P(\rho) \succeq 0.$$

The derivative term $\dot{P}$ is built with `rhodiff(P, [-1 1])`. The comparison operators create `pdlmi` wrappers, and `toYalmip` hands the coefficient-wise constraints to YALMIP. The second listing assigns the solver- and tolerance-dependent numeric value of γ to `gammaValue`. For this model, `numel(E1.Constraints)` returns 6, and `numel(E2.Constraints)` returns 2.

Example

```
yalmip('clear')
grid = linspace(0, 1, 2);
A = pdmat(grid, @(x) [-1, 0.5; -1, -2] + ...
    x * [-1.3, -20; 2, -10], Degree=1);
B = pdmat(grid, @(x) [1, -4; -1, -1] + ...
    x * [2.2, 0.5; -6, -5], Degree=1);
Cmat = eye(2);
Dmat = zeros(2);

P = pdvar(2, grid);
diffP = rhodiff(P, [-1 1]);
gamma = pdvar(1, grid, Degree=0);
E1 = [diffP + P * A + A' * P, P * B, Cmat';
      B' * P, -gamma * eye(2), Dmat';
      Cmat, Dmat, -gamma * eye(2)] <= 0;
E2 = P >= 0;
decayConstraintCount = numel(E1.Constraints)
positivityConstraintCount = numel(E2.Constraints)
```

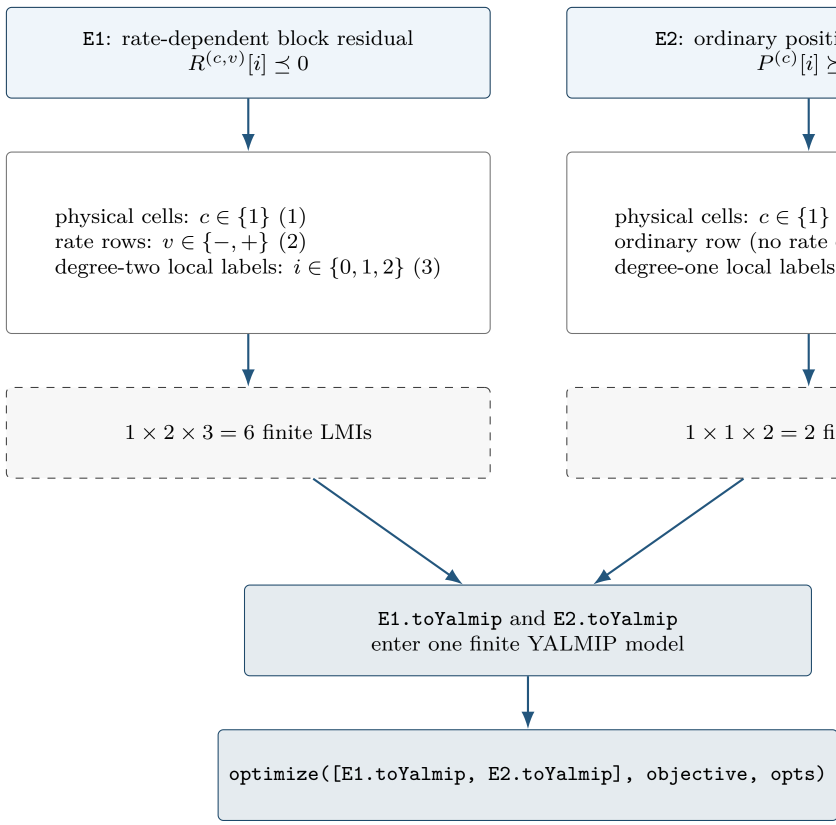
```
decayConstraintCount =
     6

positivityConstraintCount =
     2
```

Example (continued)

```
objective = gamma.LocalValues{1}{1};
solver = 'lmilab';
if exist('mosekopt', 'file') ~= 0
    solver = 'mosek';
end
opts = sdpsettings('solver', solver, 'verbose', 0);
sol = optimize([E1.toYalmip, E2.toYalmip], objective, opts);
assert(sol.problem == 0, sol.info)
gammaValue = value(objective);
assert(isfinite(gammaValue))
gammaValue
```

The constraint counts above follow directly from the three independent assembly axes: physical-cell identity, rate-row identity, and local Bernstein label. In this tested example the two wrappers use the same physical grid but different rate-row and degree structures.



The numbers (6) and (2) are certificate-assembly counts: they state how many finite semidefinite constraints represent `E1` and `E2`. They are not evidence that the optimization succeeds. Feasibility is checked separately by `sol.problem`, and the finite objective is checked after the shared `optimize` call.

A

Bernstein Basis Mathematics

This appendix develops the representation used by the package from the one-dimensional degree-one case to cell-local tensor polynomials. The reader needs only scalar polynomials, matrices, and elementary calculus. Every conversion, elevation, product, derivative, and subdivision below is exact; none is a sampling approximation. Farouki gives a broader historical and numerical account [1].

Each main construction follows the same reading path: *Basic idea*, *Formula*, *Worked example*, and *Visual conclusion*. The extra staging is intentional: first identify what the operation means, then read its indices, and only afterward connect it to package storage.

A.1 Convex Combinations Before Polynomials

Basic idea. A convex combination of numbers or matrices has the form

$$X = \sum_{i=0}^{m} w_i X_i, \qquad w_i \geq 0, \qquad \sum_{i=0}^{m} w_i = 1.$$

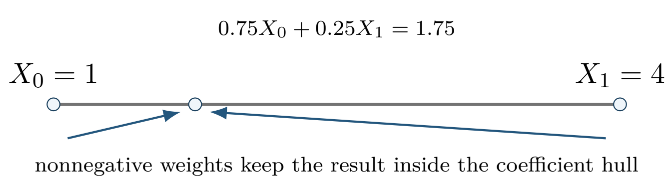
For scalars, X lies between the smallest and largest X_i. For symmetric matrices, if every $X_i \succeq 0$, then $X \succeq 0$, because

$$z^\mathsf{T} X z = \sum_i w_i \, z^\mathsf{T} X_i z \geq 0 \quad \text{for every } z.$$

Formula. Bernstein bases are useful because their basis values are precisely such weights at every point of a cell.

Worked example. Take $X_0 = 1$, $X_1 = 4$, and weights 0.75 and 0.25. Their combination is 1.75, which remains between the two inputs.

Visual conclusion.



The implication is one-way. A polynomial may be positive everywhere while one Bernstein coefficient is negative; Appendix B uses that fact to motivate stronger certificates.

A.2 Degree-One Interpolation On One Physical Cell

Basic idea. Let a physical cell be $[\rho_1^{(c)}, \rho_1^{(c+1)}]$, with width $h_c = \rho_1^{(c+1)} - \rho_1^{(c)} > 0$. Its forward local coordinate is

$$\alpha = \frac{\rho_1 - \rho_1^{(c)}}{h_c} \in [0, 1].$$

Formula. The degree-one Bernstein basis is

$$B_0^1(\alpha) = 1 - \alpha, \qquad B_1^1(\alpha) = \alpha.$$

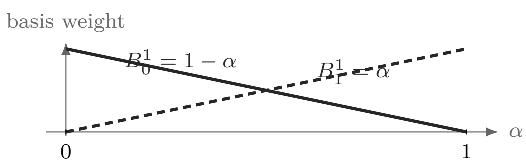
Therefore

$$M_c(\rho) = (1 - \alpha)C^{(c)}[0] + \alpha C^{(c)}[1].$$

At the lower physical face, $\alpha = 0$ and the value is $C^{(c)}[0]$; at the upper face, $\alpha = 1$ and the value is $C^{(c)}[1]$. Between them the matrix follows the straight segment joining those two coefficient matrices.

Worked example. For example, $C^{(c)}[0] = 2$ and $C^{(c)}[1] = 4$ give $M_c(0.25) = 0.75(2) + 0.25(4) = 2.5$, exactly as in section 4.4.

Visual conclusion.



A.3 Higher Degree And The Partition Of Unity

Basic idea. Higher degree adds interior shape controls while keeping the same physical cell. It does not add physical grid nodes.

Formula. For degree m, define

$$B_i^m(\alpha) = \binom{m}{i}(1 - \alpha)^{m-i}\alpha^i, \qquad i = 0, \dots, m.$$

Each basis function is nonnegative on $[0, 1]$. The binomial theorem gives

$$\sum_{i=0}^{m} B_i^m(\alpha) = ((1 - \alpha) + \alpha)^m = 1.$$

Thus

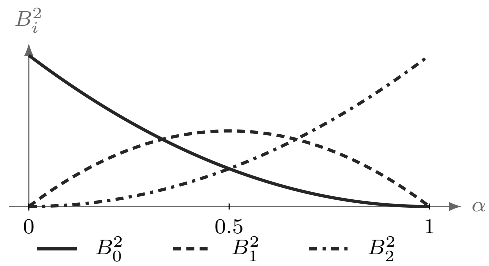
$$M_c(\alpha) = \sum_{i=0}^{m} B_i^m(\alpha)C^{(c)}[i]$$

is a convex combination of its local coefficient matrices.

Worked example. For degree two,

$$B_0^2 = (1 - \alpha)^2, \qquad B_1^2 = 2(1 - \alpha)\alpha, \qquad B_2^2 = \alpha^2.$$

Visual conclusion.



At every α, the three plotted heights are nonnegative and sum to one.

The label i identifies a basis position inside the selected physical cell. It is not a physical grid-node subscript.

A.4 Worked Conversion: $1 + 2\alpha + 3\alpha^2$

Consider

$$p(\alpha) = 1 + 2\alpha + 3\alpha^2.$$

Seek degree-two Bernstein coefficients b_0, b_1, b_2:

$$p = b_0(1-\alpha)^2 + 2b_1(1-\alpha)\alpha + b_2\alpha^2.$$

Matching powers of α gives

$$b_0 = 1, \qquad b_1 = 2, \qquad b_2 = 6.$$

Indeed,

$$1B_0^2 + 2B_1^2 + 6B_2^2 = (1 - 2\alpha + \alpha^2) + (4\alpha - 4\alpha^2) + 6\alpha^2 = 1 + 2\alpha + 3\alpha^2.$$

The three values $1, 2, 6$ are coefficients of basis functions, not samples at three equally spaced physical points. Only the two endpoint coefficients are endpoint values. At $\alpha = 1/2$,

$$p(1/2) = \tfrac{1}{4}(1) + \tfrac{1}{2}(2) + \tfrac{1}{4}(6) = 2.75.$$

Example

```
p = pdmat({[0 1]}, {1, 2, 6}, Degree=2);
valueAtHalf = p.evaluate(0.5)
T = bernsteinTable(p, "oneLine")
```

```
valueAtHalf =
    2.7500
T =
    CellSubscript                    Expression
    -------------    ------------------------------------------

        {[1]}        "(1-alpha)^2*1 + 2(1-alpha)alpha*2 + alpha^2*6"
```

A.5 Exact Power–Bernstein Conversion

More generally, use brackets for algebraic coefficient labels as well:

$$p(\alpha) = \sum_{j=0}^{n} a[j]\alpha^j = \sum_{k=0}^{n} b[k]B_k^n(\alpha).$$

Power coefficients convert to Bernstein coefficients by

$$b[k] = \sum_{j=0}^{k} \frac{\binom{k}{j}}{\binom{n}{j}} a[j], \qquad k = 0, \ldots, n.$$

The inverse map is

$$a[j] = \binom{n}{j} \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} b[i], \qquad j = 0, \ldots, n.$$

These are changes of basis inside the same degree-n polynomial space. For matrix polynomials they act entrywise. For tensor polynomials they act along one coordinate axis at a time.

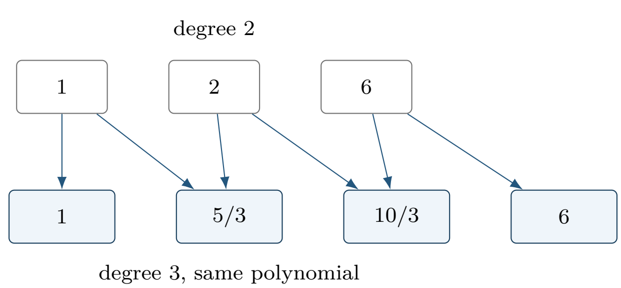
A.6 Lossless Degree Elevation

In one parameter, a polynomial on physical cell c with degree m has a Bernstein representation of every higher degree. One elevation step gives

$$\widehat{C}^{(c)}[0] = C^{(c)}[0], \qquad \widehat{C}^{(c)}[k] = \frac{k}{m+1} C^{(c)}[k-1] + \left(1 - \frac{k}{m+1}\right) C^{(c)}[k], \qquad \widehat{C}^{(c)}[m+1] = C^{(c)}[m].$$

For the worked coefficients $[1, 2, 6]$, elevation from degree two to degree three produces

$$\left[1, \frac{5}{3}, \frac{10}{3}, 6\right].$$



Direct elevation from m to $M \geq m$ is

$$\widehat{C}^{(c)}[k] = \sum_{i=\max(0,k-M+m)}^{\min(m,k)} C^{(c)}[i] \frac{\binom{m}{i}\binom{M-m}{k-i}}{\binom{M}{k}}.$$

The new coefficients are fixed convex combinations of the old ones. Elevation therefore changes representation but adds no decision freedom. Constructing a new higher-degree `pdvar` does enlarge the decision parameterization.

A.7 Physical Tensor Grids And Local Tensor Labels

Suppose direction s has physical nodes

$$\rho_s^{(1)} < \rho_s^{(2)} < \cdots < \rho_s^{(k_s)}.$$

The counts k_s may differ. Thus $\mathcal{G} = \mathcal{G}_1 \times \cdots \times \mathcal{G}_\ell$ has $\prod_s k_s$ physical nodes. A physical hypercube is identified separately by the one-based multi-index $\mathbf{c} = (c_1, \ldots, c_\ell)$. On that hypercube,

$$\alpha_s = \frac{\rho_s - \rho_s^{(c_s)}}{\rho_s^{(c_s+1)} - \rho_s^{(c_s)}}, \qquad s = 1, \ldots, \ell.$$

For the package's scalar degree m,

$$C^{(\mathbf{c})}(\boldsymbol{\rho}) = \sum_{\mathbf{i} \in \{0,\ldots,m\}^\ell} B_{\mathbf{i}}^m(\boldsymbol{\alpha}) C^{(\mathbf{c})}[\mathbf{i}], \qquad B_{\mathbf{i}}^m = \prod_{s=1}^{\ell} B_{i_s}^m(\alpha_s).$$

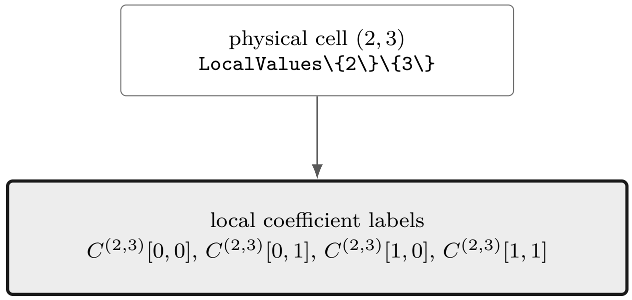
There are $\prod_s (k_s - 1)$ physical hypercubes and $(m+1)^\ell$ zero-based local Bernstein labels per hypercube. Physical node and hypercube indices remain one-based.

Example

```
obj = pdbase({[0 1 2], [10 20 30 40]}, [1 1], 1);
physicalCellCount = obj.ncell()
coefficientsPerCell = obj.ncoeff()
localLabels = obj.lbls()
```

```
physicalCellCount =
     6
coefficientsPerCell =
     4
localLabels =
     0     0
     0     1
     1     0
     1     1
```

The nested tree `LocalValues\{c1\}\{c2\}...\{cell\}` follows physical-cell coordinates. Only after reaching a leaf does the flat coefficient cell follow the `lbls` order. Earlier dimensions vary more slowly in both `cells` and `lbls`.



Select the physical-cell leaf first; only then read the flat local coefficient labels in `lbls` *order.*

A.8 Products As Ordered Weighted Convolution

Basic idea. Multiplying two Bernstein polynomials creates every ordered pair of input coefficients. Pairs with the same label sum contribute to the same output. For matrix payloads, "ordered" matters: the left factor stays on the left.

Formula. First consider one parameter and fix the one-based physical cell c. Let

$$P = \sum_{i=0}^{m} B_i^m P^{(c)}[i], \qquad Q = \sum_{j=0}^{n} B_j^n Q^{(c)}[j].$$

The basis-product identity

$$B_i^m B_j^n = \frac{\binom{m}{i}\binom{n}{j}}{\binom{m+n}{i+j}} B_{i+j}^{m+n}$$

gives

$$(PQ)^{(c)}[k] = \sum_{i+j=k} P^{(c)}[i] Q^{(c)}[j] \frac{\binom{m}{i}\binom{n}{j}}{\binom{m+n}{k}}.$$

This is an ordered matrix convolution. In general $P^{(c)}[i]Q^{(c)}[j] \neq Q^{(c)}[j]P^{(c)}[i]$.

Worked example. Take quadratic coefficients $P = [1, 2, 6]$ and affine coefficients $Q = [2, 4]$. The anti-diagonals give

$$
\begin{aligned}
R[0] &= 1(2) = 2,\\
R[1] &= (1(4) + 2\,2(2))/3 = 4,\\
R[2] &= (2\,2(4) + 6(2))/3 = 28/3,\\
R[3] &= 6(4) = 24.
\end{aligned}
$$

Example

```
P = pdmat({[0 1]}, {1, 2, 6}, Degree=2);
Q = pdmat({[0 1]}, {2, 4}, Degree=1);
R = P * Q;
productDegree = R.Degree
productCoefficients = cell2mat(R.coeffs(1))
```

```
productDegree =
     3
productCoefficients =
    2.0000    4.0000    9.3333   24.0000
```

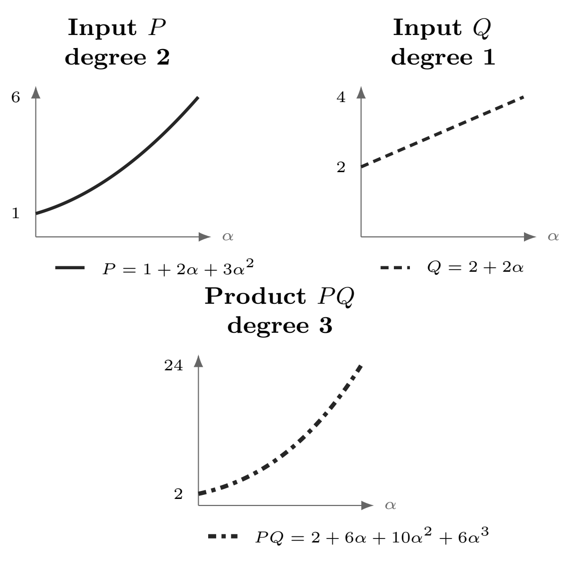
The product degree in the transcript is three: degrees add, so $2 + 1 = 3$. The four outputs are coefficients in that cubic Bernstein basis, not values sampled at four points.

The same calculation can be checked as a function identity. On $\alpha \in [0, 1]$,

$$P(\alpha) = 1 + 2\alpha + 3\alpha^2, \qquad Q(\alpha) = 2 + 2\alpha,$$

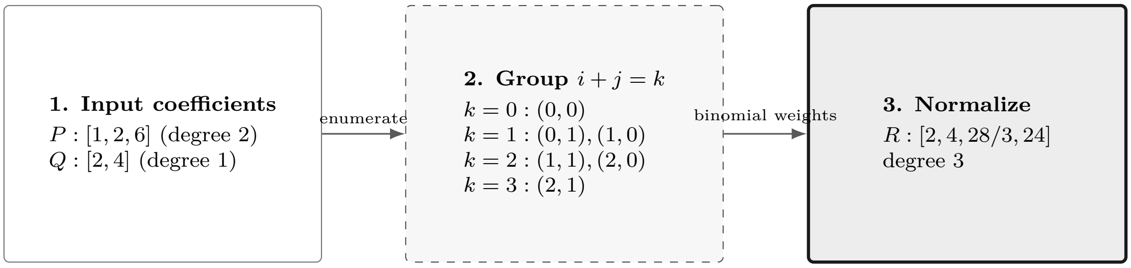
and

$$P(\alpha)Q(\alpha) = 2 + 6\alpha + 10\alpha^2 + 6\alpha^3.$$



The line samples and formulas form a legend below the three plot regions. Solid, dashed, and dash–dot strokes distinguish the functions without color; no label lies on a curve.

Visual conclusion.



Only the panel-to-panel arrows encode flow. Contributions are listed inside their anti-diagonal group, so no edges cross labels or equations.

This is the algorithm behind protected `bernProd` (section 3.6.4) and public `pdmat`/`pdvar` multiplication (sections 4.6.2 and 5.6).

For tensor labels,

$$(PQ)^{(\mathbf{c})}[\mathbf{k}] = \sum_{\mathbf{i}+\mathbf{j}=\mathbf{k}} P^{(\mathbf{c})}[\mathbf{i}]Q^{(\mathbf{c})}[\mathbf{j}] \prod_{s=1}^{\ell} \frac{\binom{m}{i_s}\binom{n}{j_s}}{\binom{m+n}{k_s}}.$$

Labels add componentwise; physical-cell identity does not change.

A.9 Differentiation And Rate Vertices

Basic idea. Differentiate inside each physical cell. Adjacent control differences produce the derivative coefficients; multiplying by a bounded rate then creates one stored row per rate-box vertex. Value continuity of P does not force the cell-local derivatives to match across a face.

Formula. For one parameter, on physical cell c of width h_c,

$$\frac{\partial P}{\partial \rho} = \frac{m}{h_c} \sum_{i=0}^{m-1} (C^{(c)}[i+1] - C^{(c)}[i]) B_i^{m-1}(\alpha).$$

Worked example. For the worked polynomial $[1, 2, 6]$ on $[0, 2]$, $m = 2$ and $h = 2$, so the derivative coefficients are

$$\left[\frac{2}{2}(2-1), \frac{2}{2}(6-2)\right] = [1, 4].$$

This represents $dp/d\rho = 1 + 3\alpha$.

In ℓ dimensions, differentiation with respect to ρ_s replaces each coefficient by

$$\frac{m}{h_s}\left(C^{(\mathbf{c})}[\mathbf{i} + \mathbf{e}_s] - C^{(\mathbf{c})}[\mathbf{i}]\right),$$

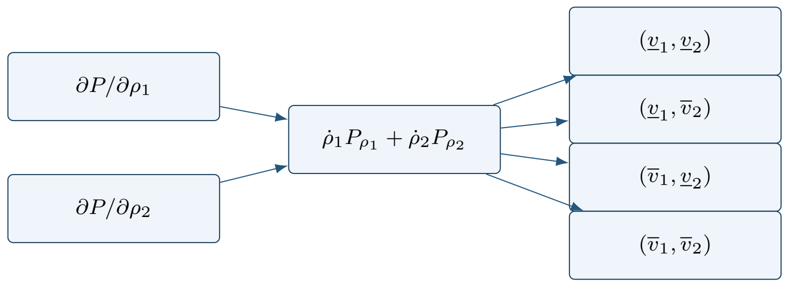
reduces the degree along direction s, and leaves the other label components unchanged. The implementation elevates partial derivatives to a common tensor degree before adding them.

Finally,

$$\dot{P} = \sum_{s=1}^{\ell} \frac{\partial P}{\partial \rho_s} \dot{\rho}_s.$$

This matrix expression is affine in $\dot{\boldsymbol{\rho}}$. Every value inside the rate box is therefore a convex combination of the 2^ℓ vertex matrices. Because the positive- and negative-semidefinite cones are convex, the requested semidefinite sign holds on the whole rate box whenever it holds at every vertex. `rhodiff` stores one row for each vertex.

Visual conclusion.



A.10 de Casteljau Evaluation And Exact Subdivision

de Casteljau recursion evaluates a Bernstein polynomial using only affine combinations. For one parameter, on a fixed physical cell c, begin with $C^{(c,0)}[i] = C^{(c)}[i]$ and define

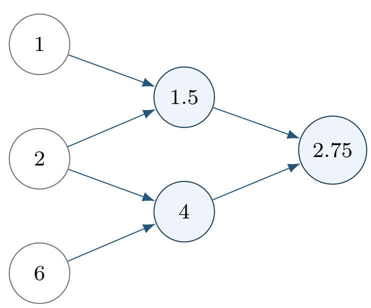
$$C^{(c,r)}[i] = (1-\tau)C^{(c,r-1)}[i] + \tau C^{(c,r-1)}[i+1], \qquad r = 1, \ldots, m.$$

The value at $\alpha = \tau$ is $C^{(c,m)}[0]$.

For $[1, 2, 6]$ and $\tau = 1/2$, the recursion is

$$[1, 2, 6] \ \longrightarrow \ [1.5, 4] \ \longrightarrow \ [2.75].$$

The left edge of this triangle gives exact coefficients $[1, 1.5, 2.75]$ on the left subinterval; the right edge gives $[2.75, 4, 6]$ on the right.



One recursion triangle supplies both the evaluation and the two child-cell coefficient lists.

Tensor evaluation or subdivision applies the same recursion along one parameter direction at a time. Subdivision changes the physical-cell partition and can tighten coefficient hulls while preserving the polynomial. The package uses exact coefficient transformations internally where documented, but it does not currently expose a public de Casteljau, subdivision, or adaptive-refinement API.

A.11 What Changes, And What Does Not

The following distinctions prevent several common modeling mistakes.

Operation	Changes	Preserves
Basis conversion	coefficient coordinates	polynomial, degree, physical cell
Degree elevation	Bernstein degree and coefficient list	polynomial, physical grid, decision freedom
Grid subdivision	physical-cell partition and local coefficients	polynomial on the original domain
Higher-degree `pdvar`	number of decision coefficients	grid and matrix size
Multiplication	polynomial degree and coefficients	physical-cell identity
Differentiation	degree and coefficient values	physical-cell partition before rate rows

These representation facts support the finite certificates in the next appendix. They do not by themselves select a solver or prove that one certificate is necessary.

B

Polynomial Matrix Sign Certificates On A Hypercube

This appendix starts with positive semidefinite matrices, builds SOS and Gram representations, and then explains the four certificate choices implemented by `pdlmi`. Background constructions are included so a new reader can recognize the mathematical landscape; only explicitly marked methods are package APIs.

As in the preceding appendix, each major certificate is revealed as *Basic idea* $\to$ *Formula* $\to$ *Worked example* $\to$ *Visual conclusion*. Keep two questions separate while reading: what polynomial matrix is represented, and what finite cone is being used to certify its sign?

For a residual $F \preceq 0$, define $S = -F$. For $F \succeq 0$, define $S = F$. The common problem is

$$S(\boldsymbol{\alpha}) \succeq 0 \quad \text{for every } \boldsymbol{\alpha} \in [0,1]^{\ell}.$$

All Gram constraints below prove non-strict inequalities. A strict model must state its own margin, for example $S - \varepsilon I \succeq 0$. The package adds no hidden margin.

B.1 Positive Semidefinite Matrices And LMIs

Basic idea. A real symmetric matrix Q is positive semidefinite when

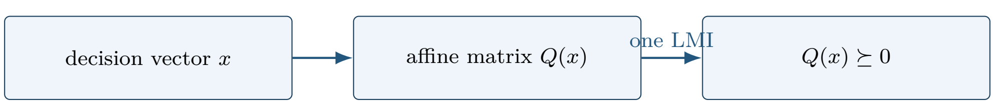
$$z^{\mathsf{T}} Q z \geq 0 \qquad \text{for every vector } z.$$

Formula. An LMI is an affine matrix expression constrained to the PSD cone:

$$Q(x) = Q_0 + x_1 Q_1 + \cdots + x_r Q_r \succeq 0.$$

The unknowns enter affinely, so this is a semidefinite program. A polynomial matrix sign condition appears infinite because it asks for one PSD matrix at every point. A certificate replaces those pointwise conditions by finitely many PSD matrices and affine coefficient identities.

Visual conclusion.



B.2 SOS And Gram Matrices

Basic idea. A scalar polynomial is a sum of squares (SOS) if

$$p(z) = q_1(z)^2 + \cdots + q_s(z)^2.$$

Every SOS polynomial is nonnegative. Choose a polynomial basis vector $v_d(z)$. The Gram form

$$p(z) = v_d(z)^\mathsf{T} Q v_d(z), \qquad Q \succeq 0,$$

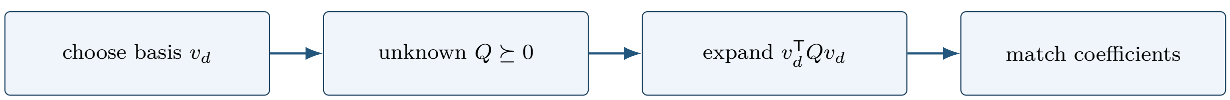
is SOS because a factorization $Q = L^\mathsf{T} L$ yields $p = \|Lv_d\|_2^2$. Expanding the Gram form and matching every coefficient of p gives equations linear in the entries of Q. Thus an SOS search is a finite PSD constraint plus affine equalities.

Formula. For a symmetric n-by-n polynomial matrix, set

$$Z_d(z) = v_d(z) \otimes I_n, \qquad S(z) = Z_d(z)^\mathsf{T} Q Z_d(z), \quad Q \succeq 0.$$

Then $y^\mathsf{T} S(z) y = (Z_d y)^\mathsf{T} Q (Z_d y) \geq 0$ for every y, so $S(z) \succeq 0$. Matrix-SOS relaxations are developed in [9].

Visual conclusion.



B.3 Full-Space SOS

An unweighted Gram form proves $S(z) \succeq 0$ on all of $\mathbb{R}^\ell$. That can be much stronger than positivity only on a box. For example, $\alpha(1-\alpha)$ is nonnegative on $[0,1]$ but negative outside it. The package does not expose a general full-space SOS parser; this construction is background for understanding weighted certificates.

B.4 Direct Bernstein Coefficients

Basic idea. If every local coefficient matrix has the requested sign, every convex combination of those coefficients has the same sign. This yields a small, transparent sufficient certificate.

Formula. On physical cell $\mathbf{c}$ and one rate row, write

$$S(\boldsymbol{\alpha}) = \sum_{\mathbf{i}} B_{\mathbf{i}}^m(\boldsymbol{\alpha}) C^{(\mathbf{c})}[\mathbf{i}].$$

Because the basis weights are nonnegative and sum to one,

$$C^{(\mathbf{c})}[\mathbf{i}] \succeq 0 \ \forall \mathbf{i} \quad \Longrightarrow \quad S(\boldsymbol{\alpha}) \succeq 0.$$

This is the package default. It introduces no Gram variables and creates one constraint per physical cell, rate row, and local label. The converse is false.

Worked example. Consider

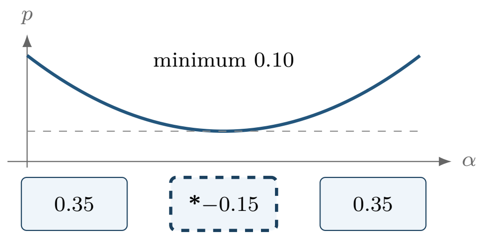
$$p(\alpha) = \alpha^2 - \alpha + 0.35 = (\alpha - 0.5)^2 + 0.10.$$

It is strictly positive, with minimum 0.10, but its degree-two Bernstein coefficients are

$$[\,0.35,\ -0.15,\ 0.35\,].$$

The negative middle coefficient defeats the direct certificate without making the polynomial negative.

Visual conclusion.



One negative Bernstein coefficient does not imply a negative curve.

The same polynomial has the PSD Bernstein Gram representation

$$\zeta_1 = \begin{bmatrix} 1-\alpha \\ \alpha \end{bmatrix}, \qquad Q = \begin{bmatrix} 0.35 & -0.15 \\ -0.15 & 0.35 \end{bmatrix} \succeq 0, \qquad p = \zeta_1^{\mathsf{T}} Q \zeta_1.$$

The eigenvalues of Q are 0.20 and 0.50. A complete Gram certificate can therefore succeed when coefficient-wise signs fail.

B.5 Pólya As Lossless Elevation

Basic idea. Pólya selection does not change the residual polynomial. It rewrites that same polynomial at a higher Bernstein degree and then retries the direct coefficient test.

Formula. Classical Pólya theory concerns homogeneous polynomials strictly positive on a simplex [12]. For one box coordinate,

$$\lambda_0 = 1 - \alpha, \quad \lambda_1 = \alpha, \quad \lambda_0 + \lambda_1 = 1.$$

For ℓ coordinates, the implemented multiplier is

$$\prod_{s=1}^{\ell} (\lambda_{s,0} + \lambda_{s,1})^d = 1.$$

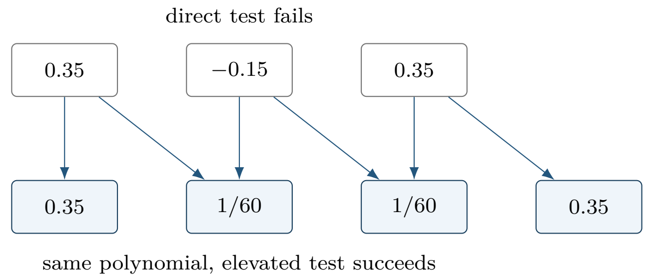
Regrouping is exactly Bernstein degree elevation: the represented residual does not change.

Worked example. For the worked coefficients,

$$[\,0.35, -0.15, 0.35\,] \longrightarrow \left[\,0.35, \frac{1}{60}, \frac{1}{60}, 0.35\,\right]$$

after one elevation. Every new coefficient is positive, so the elevated coefficient test succeeds.

Visual conclusion.



`applyPolya(d)` applies one scalar increment in every parameter direction and then coefficient constraints to every cell and rate row. It adds neither Gram variables nor decision freedom. A fixed-level failure is inconclusive; the package makes no tensor-box completeness claim.

B.6 Exact Interval Markov–Lukács Forms

Basic idea. On one interval, endpoint factors can encode the domain exactly. Even and odd target degrees require different weighted Gram patterns, so parity is a structural choice rather than a cosmetic one.

Formula. In one variable, interval nonnegativity has exact parity-specific descriptions [6, 7]. If $\deg p \le 2r$,

$$p \ge 0 \text{ on } [0,1] \iff p = s_0 + \alpha(1-\alpha)s_1,$$

where s_0, s_1 are SOS of degrees at most $2r$ and $2r-2$. If $\deg p \le 2r+1$,

$$p \ge 0 \text{ on } [0,1] \iff p = (1-\alpha)s_L + \alpha s_U,$$

where s_L, s_U are SOS of degree at most $2r$.

For polynomial matrices, with

$$\mathcal{Z}_d(\alpha) = \begin{bmatrix} B_0^d I_n & \cdots & B_d^d I_n \end{bmatrix}^{\mathsf{T}},$$

the even and odd forms are

$$S = \mathcal{Z}_r^{\mathsf{T}} Q_0 \mathcal{Z}_r + \alpha(1-\alpha)\mathcal{Z}_{r-1}^{\mathsf{T}} Q_1 \mathcal{Z}_{r-1}, \qquad Q_0, Q_1 \succeq 0,$$

and

$$S = (1-\alpha)\mathcal{Z}_r^{\mathsf{T}} Q_L \mathcal{Z}_r + \alpha \mathcal{Z}_r^{\mathsf{T}} Q_U \mathcal{Z}_r, \qquad Q_L, Q_U \succeq 0.$$

The second even block is absent at $r = 0$. See [8, 9] for polynomial-matrix context.

Worked example. A cubic residual has degree $2r+1$ with $r = 1$. It therefore uses two degree-one Gram bases: one multiplied by $1-\alpha$ and one by α. This is the one-dimensional full-box pattern used for the cubic residual in the introductory DPD-LMI example.

Visual conclusion.

even $2r$
unweighted Gram + $\alpha(1-\alpha)$-weighted Gram

odd $2r+1$
$(1-\alpha)$ lower-face Gram + α upper-face Gram

Partition Q into n-by-n blocks Q_{ij}. The Bernstein coefficient at label k of $\alpha^a(1-\alpha)^b\mathcal{Z}_d^\top Q\mathcal{Z}_d$, expressed at total degree $2d+a+b$, is

$$\mathcal{G}_k^{d,a,b}(Q) = \frac{1}{\binom{2d+a+b}{k}} \sum_{\substack{0\le i,j\le d \\ i+j=k-a}} \binom{d}{i}\binom{d}{j} Q_{ij}.$$

Equating this affine map to each target coefficient gives exact linear identities; $Q \succeq 0$ supplies the LMIs. For one parameter, `applyFullBoxPreorder` implements these exact parity forms. Its minimum absolute order is $\lfloor m/2 \rfloor$.

B.7 Putinar Modules And Schmüdgen Preorderings

For

$$K = \{z : g_1(z) \ge 0, \ldots, g_q(z) \ge 0\},$$

a Putinar quadratic module has the form

$$\sigma_0 + \sum_{j=1}^{q} \sigma_j g_j, \qquad \sigma_j \text{ SOS}.$$

Under the Archimedean hypothesis, every polynomial strictly positive on K has such a representation at some degree [10]. Compactness alone is not that theorem statement. The package implements one box-specific, fixed-order selector, not a general-domain or general-generator parser.

For one parameter, `applyPutinar(r)` dispatches to the same exact even/odd Markov–Lukács forms as `applyFullBoxPreorder(r)`. Its absolute default and minimum are $r = \lfloor m/2 \rfloor$. Even targets use the unweighted and $\alpha(1-\alpha)$-weighted Gram blocks and match at degree $2r$; odd targets use the two endpoint-weighted Gram blocks and match at degree $2r+1$.

For $\ell \ge 2$, use the local box generators

$$g_s(\boldsymbol{\alpha}) = \alpha_s(1-\alpha_s),$$

and `applyPutinar(r)` uses

$$F = S_0 + \sum_{s=1}^{\ell} g_s S_s, \qquad S_0 = \mathcal{Z}_{r\mathbf{1}}^\top Q_0 \mathcal{Z}_{r\mathbf{1}}, \quad S_s = \mathcal{Z}_{r\mathbf{1}-\boldsymbol{e}_s}^\top Q_s \mathcal{Z}_{r\mathbf{1}-\boldsymbol{e}_s},$$

where every present $Q_s \succeq 0$. In this multivariate branch the order is absolute and satisfies $r \ge \lceil m/2 \rceil$. Exact identities match the sign-normalized target, elevated without changing it, at tensor degree $2r$. Negative-degree multiplier blocks are omitted at $r = 0$. Each physical cell and active rate row owns independent Gram and equality blocks; no generator products, implicit margin, or solver call are introduced.

A Schmüdgen full preordering includes every square-free generator product:

$$p = \sum_{\boldsymbol{\nu} \in \{0,1\}^q} \sigma_{\boldsymbol{\nu}} \prod_{j=1}^{q} g_j^{\nu_j}, \qquad \sigma_{\boldsymbol{\nu}} \text{ SOS.}$$

Strict positivity on a compact basic closed semialgebraic set has such a representation [11]. A finite truncation is still a sufficient certificate, and up to 2^q multiplier blocks may be required.

B.8 The Implemented Multivariate Full Box Preordering

Basic idea. For several parameters, use every square-free product of the box generators. Each subset selects one weighted Gram block, and exact coefficient identities make the sum equal the stored residual on one physical cell and rate row.

Formula. For the hypercube, use

$$g_s(\boldsymbol{\alpha}) = \alpha_s(1 - \alpha_s), \qquad s = 1, \dots, \ell.$$

Worked example. For two parameters, the four subset products are $1, g_1, g_2, g_1 g_2$.

Visual conclusion.

$J = \varnothing$ 1	$J = \{1\}$ $\alpha_1(1-\alpha_1)$
$J = \{2\}$ $\alpha_2(1-\alpha_2)$	$J = \{1,2\}$ $g_1 g_2$

For $\ell \geq 2$ and absolute order r, define

$$\mathcal{Z}_{\boldsymbol{d}} = \left(\bigotimes_{s=1}^{\ell} \begin{bmatrix} B_0^{d_s} & \cdots & B_{d_s}^{d_s} \end{bmatrix}^{\mathsf{T}} \right) \otimes I_n.$$

The implemented form is

$$S = \sum_{J \subseteq \{1,\dots,\ell\}} \left(\prod_{s \in J} g_s \right) \mathcal{Z}_{r\mathbf{1}-\boldsymbol{\chi}_J}^{\mathsf{T}} Q_J \mathcal{Z}_{r\mathbf{1}-\boldsymbol{\chi}_J}, \qquad Q_J \succeq 0.$$

Terms with negative tensor degree are omitted. A present block has dimension

$$n \prod_{s=1}^{\ell} \left(r + 1 - \mathbf{1}_{s \in J} \right).$$

Every physical cell and active rate row gets independent dense Gram blocks and exact coefficient identities. The minimum order is $\lceil m/2 \rceil$ for $\ell \geq 2$. This is a specific dense Bernstein realization of the complete box preordering, not a full-space SOS, Putinar-only, sparse, or general-domain parser. A fixed multivariate order is sufficient rather than exact.

Finite SDP example: $\ell = 2$, $m = 3$. Let $(a, b) \in [0, 1]^2$ be the local coordinates of one cell, and suppose the sign-normalized residual is

$$F(a, b) = \sum_{i=0}^{3} \sum_{k=0}^{3} B_i^3(a) B_k^3(b) F_{i,k} \succeq 0.$$

The default order is $r = \lceil 3/2 \rceil = 2$. After exact elevation to degree $(4, 4)$, the implementation introduces four independent PSD Gram matrices and enforces the identity

$$\begin{aligned} F(a, b) &= \mathcal{Z}_{(2,2)}^{\mathsf{T}} Q_{\varnothing} \mathcal{Z}_{(2,2)} + a(1-a) \mathcal{Z}_{(1,2)}^{\mathsf{T}} Q_1 \mathcal{Z}_{(1,2)} \\ &\quad + b(1-b) \mathcal{Z}_{(2,1)}^{\mathsf{T}} Q_2 \mathcal{Z}_{(2,1)} \\ &\quad + a(1-a) b(1-b) \mathcal{Z}_{(1,1)}^{\mathsf{T}} Q_{12} \mathcal{Z}_{(1,1)}, \\ Q_{\varnothing} \succeq 0, \qquad & Q_1 \succeq 0, \qquad Q_2 \succeq 0, \qquad Q_{12} \succeq 0. \end{aligned}$$

For an n-by-n residual, these blocks have dimensions $9n$, $6n$, $6n$, and $4n$. If

$$F(a, b) = \sum_{p=0}^{4} \sum_{q=0}^{4} B_p^4(a) B_q^4(b) \widetilde{F}_{p,q},$$

then $\widetilde{F}_{0,0} = F_{0,0}$ and $\widetilde{F}_{1,0} = F_{0,0}/4 + 3F_{1,0}/4$. The finite SDP has the four PSD-LMI constraints above and one exact matrix equality for each of the $5 \times 5 = 25$ elevated Bernstein labels. With $[Q]_{u,v}$ denoting the n-by-n Gram block indexed by labels u, v, two illustrative equalities are

$$\begin{aligned} F_{0,0} &= [Q_{\varnothing}]_{(0,0),(0,0)}, \\ \frac{1}{4} F_{0,0} + \frac{3}{4} F_{1,0} &= \frac{1}{2} [Q_{\varnothing}]_{(0,0),(1,0)} + \frac{1}{2} [Q_{\varnothing}]_{(1,0),(0,0)} + \frac{1}{4} [Q_1]_{(0,0),(0,0)}. \end{aligned}$$

The remaining 23 are generated by the same weighted Bernstein coefficient map. Thus the identities are affine equalities, not 25 additional LMIs. A $\preceq 0$ comparison instead matches the negatives of the elevated target coefficients.

Example

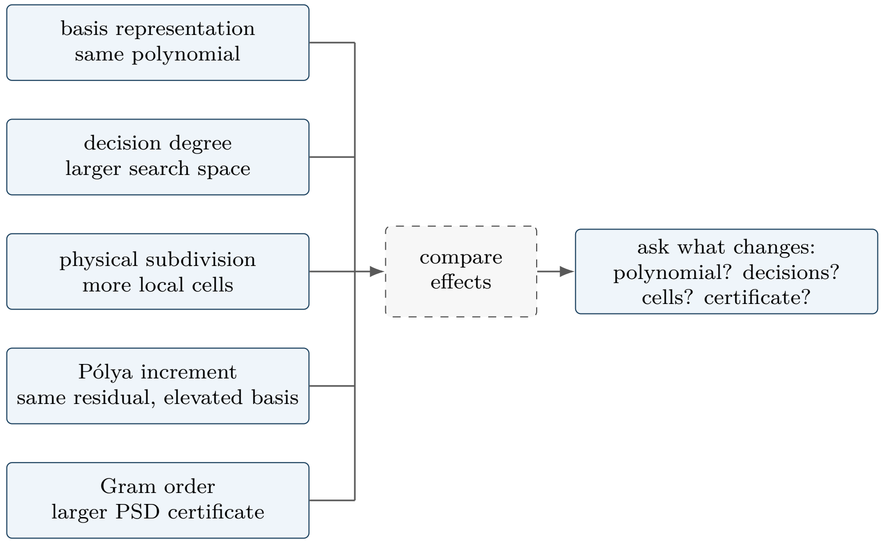
```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
polya = direct.applyPolya(1);
putinar = direct.applyPutinar();
box = direct.applyFullBoxPreorder();
directConstraintCount = numel(direct.Constraints)
polyaState = [polya.PolyaDegree, numel(polya.Constraints)]
putinarState = [putinar.PutinarOrder, numel(putinar.Constraints)]
fullBoxState = [box.FullBoxOrder, numel(box.Constraints)]
```

```
directConstraintCount =
     3
polyaState =
     1     4
putinarState =
     1     5
fullBoxState =
     1     5
```

Selections rebuild from the stored original `Residual` and `Relation`. Reapplying Pólya, Putinar, or full-box replaces the previous selection; the certificates never compound, and the three opt-in families are mutually exclusive.

B.9 Five Different Modeling Choices

These choices act on different parts of the problem.



Choice	Changes	Does not mean
basis conversion	coefficient coordinates	more decision freedom
decision degree	independent decision coefficients	certificate order
subdivision	physical cells and local hulls	higher polynomial degree
Pólya degree	elevated residual coefficients	Gram variables
Putinar/full-box order	Gram bases and PSD block sizes	decision degree

When a certificate fails, do not infer continuous infeasibility. Decide separately whether the model needs a richer decision degree, a finer physical grid, a larger Pólya increment, or a higher Putinar or full-box order.

B.10 Implemented Boundary

Implemented public behavior	Background only
direct coefficient assembly	general full-space SOS parser
uniform `applyPolya([d])`	direction-wise or automatic Pólya search
`applyPutinar([r])`	general-domain/general-generator Putinar interface
`applyFullBoxPreorder([r])`	general-domain Schmüdgen parser
exact one-dimensional parity forms	automatic or adaptive Gram-order search
all cells and active rate rows	sparse Gram selection or adaptive refinement
explicit user margins	package-owned strictness or solver policy

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